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Understanding and Predicting Micro-Donations in Washington, D.C. Political Campaigns: An Explainable Machine Learning Approach

Daniel Simões Santos

Master Thesis

presented as partial requirement for obtaining a master's degree in information management.

NOVA Information Management School

Instituto Superior de Estatística e Gestão de Informação

Universidade Nova de Lisboa

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By

Daniel Simões Santos

Master Thesis presented as partial requirement for obtaining the master's degree in information management, with a specialization in Business Intelligence.

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July, 2025

STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism, any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Lisbon, 07-07-2025

Daniel Simões Santos

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ABSTRACT

Political campaign financing has grown increasingly reliant on small-dollar micro-donations, which broaden the donor base and democratize fundraising. This thesis investigates how explainable machine learning can predict micro-donation behavior and identify its key drivers, focusing on political contributions in Washington, D.C. A comprehensive dataset of campaign finance records is compiled and integrated with socio-demographic and geographic open data to capture contextual factors and uncover spatial patterns in giving. Following the CRISP-ML framework, we develop a gradient-boosted tree model (XGBoost) to predict the amount of micro-donations contributions, using Optuna for rigorous hyperparameter tuning. To ensure transparency in this predictive approach, we apply SHAP (Shapley Additive Explanations) to interpret the models' outputs, allowing us to assess the influence of individual donor characteristics and donation attributes on the predictions. The results demonstrate high predictive accuracy, with a Mean Absolute Error (MAE) of approximately \$1.56 and an R^2 of 0.8. The results show that contributor identity is the primary driver of micro-donation behavior, followed by committee and candidate names, donation patterns, and the demographic and socioeconomic contexts. These insights highlight the strong role of individual donor behavior while also capturing the influence of organizational ties and neighborhood characteristics. Notably, the inclusion of demographic and geographic data provides valuable insights into community-level trends and patterns, enhancing our understanding of how local context shapes political giving. By combining strong predictive performance with interpretable insights, this study offers a novel, data-driven perspective on the mechanics of small-scale political contributions. The findings enrich current debates on campaign finance by highlighting the significance of micro-donations and providing evidence-based recommendations for campaign strategies and policy reforms that aim to enhance transparency, promote equity, and encourage broader civic participation in the political process.

KEYWORDS

Political Contributions; Campaign Finance; Micro-Donations; Donor Behavior; Washington, D.C.; Open Data; Machine Learning; Explainable AI (XAI); SHapley Additive exPlanations (SHAP); XGBoost; Optuna

Sustainable Development Goals (SDG):



TABLE OF CONTENTS

Statement of Integrity.....	ii
Acknowledgements.....	iii
Abstract	iv
List of Figures.....	vii
List of Tables.....	viii
List of Abbreviations and Acronyms.....	ix
1. Introduction.....	1
2. Literature Review	3
2.1 Political Contributions	3
2.2 Micro-Donations in Political Finances	5
2.3 Contributions Impact on Political Behaviour.....	6
2.4 Machine Learning Potential in the Politics Field	6
2.5 Washington, D.C., as the Political Epicenter of the United States	8
3. Methodology	9
3.1 Data.....	10
3.1.1 Study Area	10
3.1.2 Data Sources and Datasets.....	12
3.1.3 Descriptive Statistics.....	14
3.2 Methods.....	19
3.2.1 XGBoost for Predicting Political Contributions	19
3.2.2 Cross-Validation Strategy	20
3.2.3 Hyperparameter Tuning	21
3.2.4 Evaluation Metrics.....	22
3.2.5 Explainable Artificial Intelligence	24

4. Results	26
4.1 Analysis of the Predictive Model	27
4.2 Model Training and Optimization.....	27
4.3 Model Evaluation.....	31
4.4 Model Interpretability.....	32
5. Discussion	36
5.1 Identity of Contributor	36
5.2 Committee and Candidate Influence on Donors' Behavior.....	36
5.3 Amount Donated and Contributions Made per Year	38
5.4 Socioeconomic and Demographic Factors Influencing Contribution Behavior	39
5.5 Geolocation	42
5.6 Type of Contributor and Method of Contribution	42
6. Conclusions.....	44
Bibliographical References	47

LIST OF FIGURES

Figure 1 - Machine Learning Pipeline for Campaign Finance Micro-Donation Prediction	9
Figure 2 - Map of Washington, D.C. by Ward Boundaries	11
Figure 3 - Average Micro-Contributions across Ward 2.....	12
Figure 4 - Prediction error plot for campaign contribution amounts	29
Figure 5 - Residual plot for predicted campaign donation amounts	30
Figure 6 - SHAP Features impact	33
Figure 7 - SHAP Features importance	33
Figure 8 - Trends in Political Contribution Volume and Value in Ward 2 (2010 - 2024).....	38

LIST OF TABLES

Table 1 - Contribution limits in 1974.....	3
Table 2 - Alignment of Workflow with CRISP-ML.....	10
Table 3 - Political Contributions data variables, categories, and sources	14
Table 4 - Summary statistics for the target variable and features (N = 6,411).....	18
Table 5 - Optimal hyperparameters of the XGBoost model (best value and search range)....	28
Table 6 - XGBoost performance	31

LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
ACS	American Community Survey
API	Application Programming Interface
BCRA	Bipartisan Campaign Reform Act
BdP	Banco de Portugal
CMA-ES	Covariance Matrix Adaptation Evolution Strategy
CRISP-ML	Cross-Industry Standard Process for Machine Learning
DC	District of Columbia
FEC	Federal Election Commission
FECA	Federal Election Campaign Act
GDP	Gross Domestic Product
GIS	Geographic Information System
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
OLS	Ordinary Least Squares
PAC	Political Action Committee
R²	Coefficient of Determination
RMSE	Root Mean Squared Error
SHAP	SHapley Additive exPlanations
TPE	Tree-structured Parzen Estimator
VAP	Voting Age Population
XAI	Explainable AI
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION

Political campaigns rely heavily on financial contributions, which have become an increasingly pivotal element of electoral success. Over recent decades, political donations have grown significantly, positioning them as a critical factor in both campaign financing and electoral outcomes (Socker et al., 2013). Among these, micro-donations play a particularly vital role by enhancing accessibility, fostering collective action, and broadening the base of political participation. Their small-dollar nature allows for non-intrusive engagement, enabling a wider segment of the population to contribute and thereby democratizing political involvement and reducing dependence on traditional high-value donors (Rubenstein, 2022).

While this growth offers candidates new opportunities, it also raises pressing concerns regarding its influence on candidate behavior, the rise of “dark money” (Alvarez et al., 2020), and the overall fairness and transparency of electoral processes. Understanding and predicting political contributions, particularly micro-donations, is crucial for analyzing influence patterns, uncovering hidden dynamics, and examining broader implications for legislative priorities and political decision-making (Lalisse, 2022). The 2020 U.S. presidential election underscored this urgency, as campaigns reached record-breaking spending levels, highlighting the structural barriers facing candidates without substantial financial backing (Myers, 2021). In politically active regions such as Washington, D.C., where campaign contributions represent a significant source of income for congressional races (Barnes et al., 2023), comprehending the factors that shape contribution behavior is especially critical. As financial resources become increasingly linked to political influence, the ability to sustain a well-funded campaign has become a central pillar of electoral competitiveness (OECD, 2024). However, this growing reliance on funding raises fundamental ethical concerns, including transparency, equity, and the potential for corruption, thus calling for robust regulatory frameworks and ongoing reform efforts (Parvin & Saunders, 2018).

Against this backdrop, this study investigates the central research question: *“How can explainable machine learning models be used to accurately predict political micro-donations in Washington, D.C., and what are the key factors that influence these contributions?”*. The specific objectives of this research are: (1) To identify and analyze the most influential donor characteristics and donation attributes, with a focus on understanding how individual profiles and contextual factors shape micro-donation behavior; (2) To employ explainable AI techniques, specifically SHAP (SHapley Additive exPlanations), to interpret model predictions and assess the relative importance of various features influencing political micro-donations; and, (3) To evaluate the contribution of demographic and geographic open data when integrated with campaign finance records, for generating actionable insights that can inform political strategy, policy reform, and future academic research on campaign finance.

The analysis examines various aspects of political giving, including contributor identity and historical donation behavior, the influence of committees and candidates on contribution patterns, the impact of donation volume and frequency, and the role of socioeconomic, demographic, and geographic context in shaping donor behavior. With a particular emphasis on micro-donations, the study sheds light on the mechanisms of small-scale political giving and its predictive potential. These findings are expected to support ongoing debates about campaign finance regulation, the influence of dark money, and the development of data-driven strategies to enhance political engagement and transparency.

To achieve these goals, the research is grounded in the CRISP-ML (Cross-Industry Standard Process for Machine Learning) (Kolyshkina & Simoff, 2021) methodology and employs a combination of advanced machine learning tools. XGBoost, a high-performance gradient boosting algorithm, is utilized to construct robust predictive models that can handle complex, high-dimensional data. Optuna is implemented for hyperparameter optimization, ensuring that model configurations are tuned for accuracy and stability. SHAP is leveraged to explain the predictions, offering a transparent view of how individual features influence model outputs and highlighting the most impactful predictors of political contributions in Washington, D.C.

Through the integration of machine learning with explainable AI and structured methodology, this thesis delivers a comprehensive analysis of political micro-donations. It provides insights into the types, timing, and determinants of contributions, including indicators of dark money activity. Ultimately, by combining predictive power with interpretability, the study contributes to the development of evidence-based strategies for fundraising, policymaking, and regulation, offering valuable guidance for political organizations, analysts, and researchers navigating the complex and evolving landscape of political finance.

The remainder of this paper is structured as follows: Chapter 2 reviews the relevant literature. Chapter 3 provides a detailed overview of the data, methodology, and machine learning and explainable AI techniques employed. Chapter 4 presents and analyzes the results, outlining the approach taken to address the research objectives. Chapter 5 discusses the findings and their broader implications. Finally, Chapter 6 concludes with a summary of key contributions and suggestions for future research directions.

2. LITERATURE REVIEW

2.1 POLITICAL CONTRIBUTIONS

Political contributions refer to financial or material donations made to support candidates, political parties, or organizations involved in electoral campaigns. These contributions are essential for funding various campaign activities, including advertisements, staff salaries, travel expenses, events, and voter outreach efforts (Myers, 2021). Contributions are categorized into two types: cash contributions and in-kind contributions. Cash contributions involve monetary donations given directly to campaigns, while in-kind contributions include the donation of goods or services instead of money.

In the United States, the regulation of campaign contributions has undergone significant evolution over time, shaped by legislative actions and judicial rulings. The Federal Election Campaign Act (FECA) of 1971 marked a turning point as the first comprehensive attempt to regulate campaign financing. It introduced disclosure requirements and spending limits, addressing two pressing concerns: the skyrocketing costs of campaigns, which limited participation to the wealthy or well-funded, and the growing influence of money in politics, which risked corruption or the appearance of it (Abrams & Settle, 2004). While the initial enactment of FECA marked significant progress, specific limits on contributions from individuals and Political Action Committees¹ (PACs) were only established in 1974 (Table 1), triggered by the revelations of the Watergate scandal² and heightened concerns about political corruption (Issacharoff, 2010). Individuals were restricted to \$1,000 per election, while PACs could contribute up to \$5,000 to a maximum of 3 candidates (House, 1974). These limits were paired with the creation of the Federal Election Commission (FEC), an independent body tasked with enforcing campaign finance laws and ensuring transparency.

Table 1 - Contribution limits in 1974

Donor Type	Contribution per candidate
Individual	\$1.000 per election
PAC Contributions	\$5.000 per election

¹ Political Action Committees that can raise and spend unlimited amounts of money to influence elections, while remaining independent of candidates and their campaigns.

² A political crisis in the 1970s in which operatives linked to the U.S. President's re-election campaign were caught breaking into the Democratic National Committee headquarters. The subsequent cover-up efforts, abuse of power, and illegal activities exposed corruption, ultimately leading to the U.S. President's resignation.

A landmark moment in the evolution of campaign finance came in 1976 with the Supreme Court's decision in *Buckley v. Valeo*. The Court upheld contribution limits, affirming their role in curbing corruption and its appearance, but struck down expenditure limits as unconstitutional restrictions on free speech. This distinction has shaped subsequent campaign finance jurisprudence, reinforcing the notion that while independent expenditures are a free speech right, contributions can be regulated to prevent undue influence (Alschuler et al., 2017).

The 2002 Bipartisan Campaign Reform Act (BCRA) represented another major shift in campaign finance regulation, targeting the growing use of soft money³ in U.S. political campaigns, as political parties increasingly used these funds to exert disproportionate influence by channeling unlimited resources into campaigns. The BCRA banned soft money contributions to national parties and introduced periodic inflation adjustments to contribution limits (Sebold, 2013). Among its provisions, the BCRA raised the individual contribution limit from \$1,000 to \$2,000 per election, per candidate, and established an aggregate contribution cap of \$117,000 over a two-year election cycle. This adjustment was significant in ensuring that limits kept pace with the rising costs of campaigns, addressing concerns that stagnant limits could erode their regulatory impact.

Periodic inflation adjustments became an established practice in 2004, raising the individual contribution limit to \$2,100. Subsequent adjustments, implemented approximately every two years, reflected inflationary trends, with the limit increasing to \$2,900 by 2022 and reaching \$3,300 for the 2023–2024 election cycle (FEC, 2023).

Despite these measures, the influence of money in politics witnessed a significant shift with the Supreme Court's decision in *Citizens United v. FEC* (2010). This ruling allowed corporations and unions to make unlimited independent expenditures, asserting that such spending was a form of free speech. The decision laid the foundation of Super PACs, which can raise and spend unlimited amounts of money as long as they operate independently of campaigns. While soft money contributions to parties remained formally prohibited, Super PACs effectively circumvented these restrictions, raising concerns about the disproportionate influence of wealthy donors in shaping political outcomes (Cox, 2023).

Further loosening of campaign finance restrictions occurred in 2014 with the Supreme Court's decision in *McCutcheon v. FEC*, which struck down aggregate contribution limits. This ruling allowed donors to contribute to an unlimited number of candidates and committees, intensifying concerns about the concentration of political influence among affluent individuals (Krell, 2015).

³ Unregulated funds raised by political parties for activities like voter registration, party-building efforts, or general party advertising, that were not subject to federal limits.

Regarding Washington, D.C., the capital offers a unique perspective on campaign finance reform. While its approach initially mirrored federal policies, the 2018 Fair Elections Amendment Act marked a departure by establishing a public financing system for local elections. This legislation aimed to reduce reliance on large-scale donations by providing matching funds for small contributions. For instance, mayoral candidates could receive matching funds for contributions capped at \$200, while councilmember candidates were limited to \$50 contributions. These reforms not only sought to amplify the voices of small donors but also aimed to create a more equitable and transparent electoral process in the district.

Political contributions remain a dynamic and integral aspect of democratic systems, continually evolving in response to shifts in legislation, societal values, and economic contexts. As they continue to shape the political landscape, studying their patterns and impact will be crucial for fostering transparency, equity, and accountability in future campaigns and governance.

2.2 MICRO-DONATIONS IN POLITICAL FINANCES

Micro-donations have emerged as a significant force in modern political campaigns, particularly in the United States, where they have played a key role in shaping fundraising strategies and broadening donor participation. Their influence has been particularly evident since 2012, with the rise of digital fundraising platforms such as ActBlue, which alone facilitated nearly \$700 million in contributions by 2018 (Albert & La Raja, 2020). This surge in micro-level giving reflects a broader transformation in the campaign finance landscape, in which a vast number of small-dollar contributions increasingly supplement traditional reliance on large donors. The expansion of micro donations has been particularly beneficial for Democratic candidates, as it not only increases available funding but also strengthens grassroots support and enhances the demographic representativeness of the donor base (Albert & La Raja, 2020). Since 2008, the number of individuals contributing small amounts to political causes, both within and beyond the realm of electoral politics, has grown exponentially, suggesting a shift in how citizens engage financially with the political process.

From a normative perspective, micro-donations have sparked debates about their democratic value. Critics of large-scale political financing often celebrate small donations as a remedy to the disproportionate influence of wealthy elites. However, some theorists remain cautious. Rubenstein (2022) highlights that both advocates and skeptics tend to overlook the unique monetary nature of these contributions, focusing instead solely on their size. Drawing on Wendy Brown's distinction between monetization and economization, Rubenstein argues that small-money donations may possess democratic potential not merely because they are modest in size, but precisely because they are monetary. According to this view, the mobility, divisibility, commensurability, and fungibility of money render micro donations particularly suited to democratic engagement. These characteristics enable such contributions to be

accessible, non-intrusive, and inherently collective, thereby facilitating broader citizen participation. Rubenstein concludes that while money is typically seen as the medium of economic exchange, in the context of micro donations, it can also function as a currency of democratic politics.

The literature suggests that micro-donations do more than offer an alternative funding mechanism; they reframe the relationship between citizens and political influence. Their growing role in campaign finance highlights critical questions about accessibility, equity, and the redefinition of civic engagement through economic means.

2.3 CONTRIBUTIONS IMPACT ON POLITICAL BEHAVIOUR

Campaign contributions are a significant source of income for political campaigns, and the need to understand their evolution is growing significantly. Greater economic power translates into greater political power and the importance of establishing a robust financial campaign during the process of applying for a political election is increasingly recognized as a pivotal element in achieving electoral success, as contributions constitute more than 50 percent of the financial resources that members of Congress procure (Socker et al., 2013). This high percentage demonstrates the significant impact that political donors have, as expectations or implied demands often accompany their contributions. To secure continued contributions, candidates often feel compelled to adapt their political positions or adjust their rhetoric in ways that align with the objectives and beliefs of their contributors (Dunaway & Munoz-Garcia, 2020).

A theoretical framework known as the "Theory of Party Competition" proposes that political parties compete in a marketplace populated by investors whose monetary contributions are critical for securing electoral victories (Ferguson, 1995), compelling political parties to formulate a strategic approach that aligns with the preferences of their electorate while simultaneously addressing the requirements of prospective donors. Candidates are required to secure a minimum level of backing from organized financial supporters, which may manifest as widespread endorsement through entities such as labor unions.

2.4 MACHINE LEARNING POTENTIAL IN THE POLITICS FIELD

In recent years, Machine learning (ML) has emerged as a revolutionary tool, providing simple solutions to complex problems across diverse fields. Its ability to extract powerful insights from vast datasets has transformed industries, and it is now poised to make an impact in the political and financial spheres, offering unprecedented capabilities for identifying patterns among donors and predicting contribution outcomes based on historical data.

When implemented within the Cross-Industry Standard Process for Data Mining (CRISP-DM), machine learning applications follow a structured framework that encompasses business understanding, data preparation, modeling, evaluation, and deployment. This systematic

approach ensures that predictive models deliver actionable insights, aligning with the overarching goals of political campaigns and mirroring their success in the financial sector (Plotnikova et al., 2022).

Regarding the models used to solve this type of problem, regression models, which are widely recognized for their ability to identify patterns, are particularly effective in financial contexts. For instance, ML regression algorithms have been successfully used in multinational stock market studies to identify trends and project future outcomes in financial datasets (Behera et al., 2023). These capabilities translate seamlessly to political applications, where forecasting contributions and voter behaviour are key strategic objectives.

To deepen these insights, Geographic Information Systems (GIS) can be integrated into ML models by incorporating spatial data, enabling the analysis of regional contribution trends and linking donor behaviour to demographic, socioeconomic, and political factors. The synergy of ML and GIS facilitates data-driven decision-making, empowering campaigns to target specific regions and tailor outreach strategies. This combination has proven its utility in diverse fields, from modelling areas at risk of flooding under extreme weather conditions (Motta et al., 2021) to predicting real estate price trends in smart cities (Trindade Neves et al., 2024).

Among the advanced machine learning techniques available, models such as XGBoost provide significant advantages in financial applications due to their ability to handle complex datasets and improve accuracy. By employing techniques such as gradient boosting, regularization (L1/L2), and tree pruning, XGBoost achieves high predictive accuracy in financial datasets, enhancing supervisory efficiency and improving auditing processes, as demonstrated in the Financial Fraud Detection field by achieving exceptional performance, with precision of 93.17% and a recall rate of 92.23% allowing for effective early detection of fraudulent activities (Li & Xu, 2023).

The model's performance can be further enhanced using tools as Optuna, which automates hyperparameter optimization to maximize efficiency and accuracy. Optuna's impact has been demonstrated in applications like healthcare, where it enabled an 86% accuracy rate in predicting mortality outcomes for stroke patients (Tikaningsih et al., 2024). While Optuna enhances model performance, SHAP (SHapley Additive exPlanations) complements it by ensuring transparency and interpretability in predictions. SHAP plays a crucial role by calculating the contribution of each input variable to the model's output. This interpretability is essential in fields where accountability is paramount, such as politics and healthcare. SHAP not only clarifies the influence of variables but also provides visual tools for comprehending complex models (Ponce-Bobadilla et al., 2024). For example, SHAP has been instrumental in identifying risk factors for heart failure, showcasing its utility across high-stakes domains (Luo et al., 2024).

In the political field, the integration of ML models with tools such as GIS, Optuna, and SHAP offers transformative potential. By enabling precise predictions of campaign contributions and

voter dynamics, these technologies provide a data-driven foundation for strategic decision-making, thereby amplifying the understanding of modern political campaign contributions.

2.5 WASHINGTON, D.C., AS THE POLITICAL EPICENTER OF THE UNITED STATES

Officially known as the District of Columbia, Washington, D.C., is the capital of the United States, with a population of approximately 700,000. The district is renowned for its robust economy, boasting the highest GDP per capita in the country (Statista, 2023), and for being the epicenter of the political domain in the United States. It hosts the executive, legislative, and judicial departments of the federal government, serving as the focal point for national legislative and decision-making processes. Key political institutions, such as the Supreme Court, the U.S. Capitol, and the White House, make Washington, D.C., the hub of national diplomacy. Additionally, the city holds significant international influence, being home to over 175 embassies and numerous global organizations, including the World Bank and the International Monetary Fund (Mayer et al., 2017).

Politically, Washington, D.C. has an unquestionable influence on the decision-making processes that shape U.S. politics, largely due to the complex interplay of interests between federal oversight, local governance, and racial dynamics (Razavi, 2022).

Given Washington, D.C.'s pivotal role in politics and economics, analysing and predicting future campaign contributions in the district is essential for understanding patterns of political influence and electoral strategy across the United States. The research reveals a wide range of opportunities to identify patterns and key influencers of political contributions in a region where political authority and economic power converge significantly.

3. METHODOLOGY

This section presents the data and the methods used in our study to predict contributions to political campaigns in Washington, D.C., using ML and XAI.

The methodological workflow for campaign finance micro-donation prediction depicted in Figure 1 follows a structured machine learning pipeline comprising eight key phases. The process begins with the collection of campaign finance and census data, followed by data preprocessing to clean, filter, and spatially enrich the dataset, reducing the initial records to a robust analytical sample. Feature engineering is then employed to create aggregation features, composite variables, and to encode and scale the data appropriately. Model development centers on selecting and training XGBoost models to capture complex non-linear relationships in donation behavior, with hyperparameter optimization performed using Optuna's TPE algorithm. Model validation is conducted through 5-fold cross-validation, using MAE as the primary optimization metric. The evaluation framework incorporates multiple assessment metrics, including MAE, MAPE, RMSE, and R^2 , to ensure model robustness. Finally, interpretability analysis is achieved through SHAP, enabling clear explanations of model predictions and the identification of the most influential features, thereby ensuring both accuracy and transparency in the predictive modeling of political micro-donations.

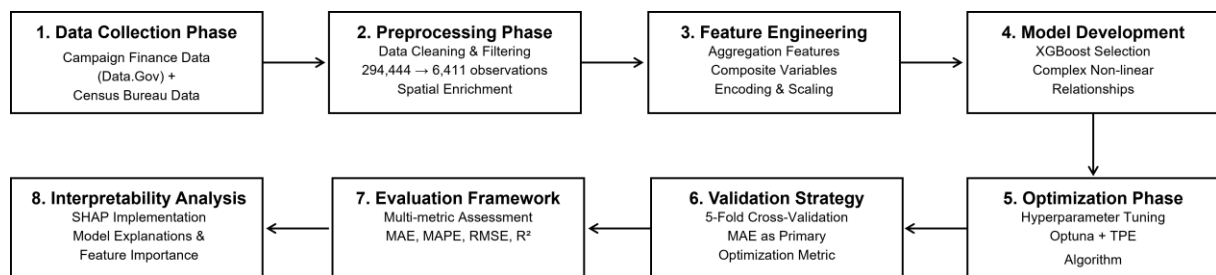


Figure 1 - Machine Learning Pipeline for Campaign Finance Micro-Donation Prediction

This methodological workflow is structured in alignment with the CRISP-ML (Cross-Industry Standard Process for Machine Learning) framework (Kolyshkina & Simoff, 2021), which provides a systematic and transparent approach to developing machine learning solutions. Each phase of the workflow, from data collection and preprocessing, through feature engineering, model development, and optimization, to validation, evaluation, and interpretability, corresponds directly to the major stages of the CRISP-ML process (Table 2). By following this framework, the study ensures methodological rigor, reproducibility, and the integration of explainability, making it possible not only to achieve accurate predictions but also to deliver clear and actionable insights for stakeholders in the context of political campaign finance.

Table 2 - Alignment of Workflow with CRISP-ML

CRISP-ML Phase	Workflow Step(s)
Business Understanding	Study objectives, research framing
Data Acquisition/Understanding	1. Data Collection, 2. Preprocessing
Data Preparation	3. Feature Engineering
Modeling	4. Model Development, 5. Optimization Phase
Evaluation	6. Validation Strategy, 7. Evaluation Framework
Explainability	8. Interpretability Analysis
Deployment & Insights	Policy recommendations

3.1 DATA

This sub-section describes the data used in this study, outlines the study area and the individual datasets used for analysis, including our selection criteria and data sources. The method of feature engineering, which involves selecting and refining dataset characteristics for maximum relevance and insight, is then outlined. We finish with descriptive statistics that describe the important features of our data, laying the groundwork for the subsequent analysis.

3.1.1 STUDY AREA

Washington, D.C., one of the most influential capitals in the world, serves as the seat of government for the United States of America and is the geographical focus of this study on political contributions. Bordering the states of Maryland and Virginia, the city holds a unique status as a federal district, independent of any state jurisdiction. According to the U.S. Census Bureau, as of July 1, 2023, Washington, D.C. had an estimated population of approximately 678,000 residents, with a population density of 4,289 inhabitants per square kilometer. However, more recent data suggests that the city's population has now surpassed 700,000, reflecting continuous demographic growth.

Politically, Washington, D.C., stands as one of the most significant hubs in the United States, a fact reinforced by the city's close engagement with political campaigns and its status as the home to the White House. Over the past 22 years, a total of 294,444 political donations have been recorded, with an average contribution of approximately \$376.67 per donation, totaling €110,910,146 in campaign contributions. These figures highlight the city's crucial role in the national landscape of political financing in the United States.

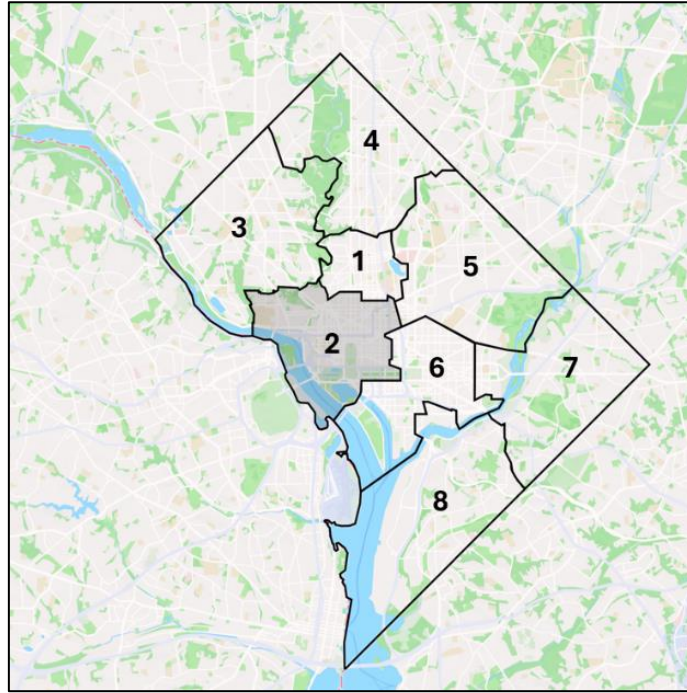


Figure 2 - Map of Washington, D.C. by Ward Boundaries

Figure 2 displays the geographic boundaries of all eight wards in Washington, D.C. This study focuses specifically on Ward 2, as it represents the ward with the highest total value of donations and is home to major political institutions, including the White House. Ward 2 spans approximately 5.9 square miles (about 15.3 km²) and includes key neighbourhoods such as Georgetown and Dupont Circle, both situated near the Potomac River, which forms the city's southwestern boundary. According to the latest estimates, Ward 2 has a population of approximately 78,900 residents, resulting in a population density of around 5,160 inhabitants per square kilometer. Its centrality, dense population, proximity to the river, and strong political engagement make Ward 2 an ideal focal point for analysing the dynamics of political contributions in the nation's capital.

Figure 3 displays the micro-contributions in Ward 2, highlighting that the highest volume of contributions is predominantly located in the northern areas of the ward.

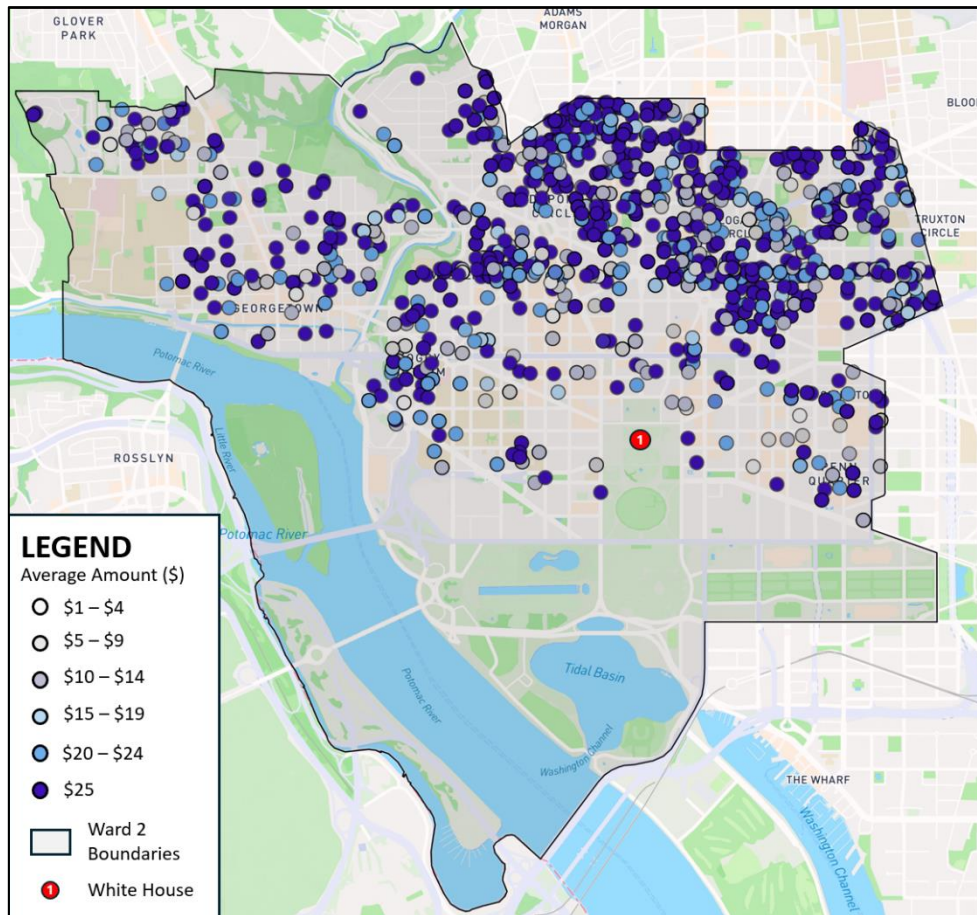


Figure 3 - Average Micro-Contributions across Ward 2

3.1.2 DATA SOURCES AND DATASETS

The dataset used in this study originates from financial contributions made to political campaigns in Washington, D.C. It provides a comprehensive record of campaign financing, capturing contributions over multiple election cycles. The dataset comprises 294,444 entries, each corresponding to a unique financial contribution. This dataset was freely downloaded from the United States Government's open data⁴.

It encompasses 22 variables, covering both intrinsic and extrinsic attributes related to financial contributions. The dataset records the names of candidates and their associated political committees receiving contributions, as well as the names, addresses, and employment information of donors. Financial information includes the contribution amount, the type of contribution (such as cash, credit card, check, or in-kind), and the date of receipt. Additionally,

⁴ Financial Contributions Dataset (<https://catalog.data.gov/dataset/campaign-financial-contributions-023f6>)

geographical and administrative attributes provide further insights, including geographical coordinates, postal addresses, and ward-level information for contributions.

Predictive modelling seeks to leverage data that enhances understanding of trends and facilitates accurate forecasting of the target variable, “Amount”.

A key aspect of this process is data enrichment, which involves augmenting existing datasets with additional contextual information to uncover deeper insights. To enrich the model, we integrated socioeconomic and demographic indicators, specifically median income, unemployment rate, education level, household size, median age, citizen voting-age population, usage of public transportation, and foreign-born population. These data were retrieved from the U.S. Census Bureau via its public API⁵, using the ACS 5-Year Estimates at the tract level for Washington, D.C. For each year from 2010 to 2024, we programmatically collected values from both the detailed tables (B-series) and subject tables (S-series), covering all tracts in the district. Each donor’s geographic coordinates, specifically the latitude and longitude associated with the contribution location, whether a residence or designated site such as a bank branch, were used to map contributions to their corresponding census tract. This spatial linkage enabled the integration of local tract-level socioeconomic indicators with each donation record, providing a more contextually enriched and nuanced analysis of political contribution behavior. Due to the absence of 2020 ACS 5-Year data (stemming from disruptions caused by the COVID-19 pandemic), we imputed missing values by averaging the results from 2019 and 2021. For 2024, preliminary estimates were generated by duplicating 2023 values as placeholders. Table 3 presents the features used in the model.

⁵ API census.gov (<https://api.census.gov/data>)

Table 3 - Political Contributions data variables, categories, and sources

Variable	Description	Data Source ⁶
ElectionYear	Year of the election.	Data.Gov
Latitude	Geographic latitude of the contribution location.	Data.Gov
Longitude	Geographic longitude of the contribution location.	Data.Gov
ContTypeXContrType	Combination of contribution type and contributor type.	Data.Gov
ContributorName	Name of the person or entity making the contribution.	Data.Gov
CandidateName	Candidate receiving the contribution.	Data.Gov
CommitteeName	Political committee receiving or associated with the	Data.Gov
DistWh	District or ward identifier (e.g., Ward number).	Feat.Eng.
QtyDonCandYr	Number of contributions to this candidate in the year.	Feat.Eng.
QtyDonCommYr	Number of contributions to this committee in the year.	Feat.Eng.
QtyDonYr	Number of contributions in the year.	Feat.Eng.
AmtCandYr	Total amount donated to this candidate in the year.	Feat.Eng.
AmtCommYr	Total amount donated to this committee in the year.	Feat.Eng.
AmtYr	Total amount donated in the year.	Feat.Eng.
AmtContTypeXContrTypeYr	Total amount donated by this contribution-contributor	Feat.Eng.
Qty25\$DonYr	Count of donations exactly \$25 in the year.	Feat.Eng.
Qty50\$DonYr	Count of donations exactly \$50 in the year.	Feat.Eng.
Qty100\$DonYr	Count of donations exactly \$100 in the year.	Feat.Eng.
Qty250\$DonYr	Count of donations exactly \$250 in the year.	Feat.Eng.
Vap	Voting age population in the associated geographic area.	Census.Gov
MedAge	Median age of residents in the associated area.	Census.Gov
MedInc	Median household income in the associated area.	Census.Gov
UnempRate	Unemployment rate in the associated area.	Census.Gov
BachelorHigher	Proportion or count of residents with a bachelor's degree	Census.Gov
AvgHhSize	Average household size in the area.	Census.Gov
ForeignBorn	Proportion or count of foreign-born residents.	Census.Gov
MedHomeVal	Median home value in the area.	Census.Gov
PubTransit	Proportion or count of public transit commuters.	Census.Gov
Amount	Individual donation amount for this record.	Data.Gov

3.1.3 DESCRIPTIVE STATISTICS

Before implementing our machine learning models, we conducted a comprehensive data preprocessing phase aimed at ensuring data integrity, reducing bias, and optimizing the feature set for modelling.

⁶ Data.gov: The U.S. Government's open data platform (<https://www.data.gov/>); Census.gov: U.S. Census Bureau (<https://www.census.gov/>)

We began by addressing missing values in the Ward column. Since the Ward variable is central to our analysis, rows lacking this information were deemed unusable and subsequently removed. This step resulted in the exclusion of 95,792 rows, leaving a total of 201,168 observations.

Next, we applied a data quality filter to the “Amount” column by removing rows with negative values (43), zero values (251), or missing values (2). This ensured that all monetary contributions in the dataset reflected valid, positive amounts, resulting in 200,872 rows.

To prevent the introduction of bias and maintain model reliability, we eliminated features with high proportions of missing data. Specifically, Employer (33.57% missing) and EmployerAddress (54.32% missing) were excluded due to their high rates of missing data. Additionally, redundant or irrelevant features such as FullAddress, Address, XCoord, YCoord, GisLastModDttm, and AddressId were excluded. These variables either duplicated spatial information or contained system-generated metadata with no analytical value. DateOfReceipt was also dropped, as its temporal inconsistencies suggested it reflected batch processing timestamps rather than actual receipt dates.

Following these exclusions, we addressed remaining missing values: 919 in ContributionType, 177 in ContributorType, 20 in ContributorName, and 15 in ElectionYear. These rows were dropped in the order listed, reducing the dataset to 199,811 rows. In contrast, we opted to impute missing values in CandidateName (30,573 missing) using the mode within grouped subsets defined by ElectionYear, CommitteeName, and Ward.

To enrich our dataset with socioeconomic and demographic features, we conducted a spatial join using Latitude and Longitude to assign each observation to its corresponding census TractId. However, because some coordinates could not be matched, and given that the socioeconomic data is only available from 2010 onward, we excluded all unmatched entries. In total, 37.62% of the rows could not be geospatially linked or lacked valid values. These were removed by filtering out missing and negative values in the MedAge, MedHomeVal, and MedInc features, resulting in a final dataset comprising 105,157 observations.

After this, we also addressed outliers in the Amount feature. Since the model was created to predict micro donations, and 20.38% of all contributions were less than or equal to \$25, we designated contributions above this threshold as outliers to be treated accordingly, resulting in a dataset of 22,408 rows. We then filtered the dataset to include only contributions made within Ward 2, resulting in a final dataset of 6,411 observations.

During preprocessing, we also engineered several new features:

- Amount CandidateName Year (AmtCandYr): Total monetary value of contributions received by each candidate in a given election year, enriching the model with target-level financial density.

- Amount CommitteeName Year (AmtCommYr): Total monetary value of contributions received by each committee in a given election year, complementing the above variable on an organizational scale.
- Amount ContributionType x ContributorType Year (AmtContTypeXContrTypeYr): A monetary aggregate capturing the total value of contributions per unique combination of contribution type and contributor type, per election year.
- Amount Year (AmtYr): Total monetary value of contributions aggregated by election year, reflecting the geographic distribution of financial support throughout the city.
- ContributionType x ContributorType (ContTypeXContrType): A composite categorical feature capturing interaction effects between the type of contribution and the type of contributor. This helps model how different donor categories interact with donation modalities.
- Donation Amount Buckets: A set of four features representing the count of contributions within specific dollar brackets per election year:
 - Qty25\$DonYr: Number of contributions of exactly \$25.
 - Qty50\$DonYr: Number of contributions of exactly \$50.
 - Qty100\$DonYr: Number of contributions of exactly \$100.
 - Qty250\$DonYr: Number of contributions of exactly \$250.
- Qty Donations CandidateName Year (QtyDonCandYr): Total number of contributions received by each candidate in a given election year, allowing the model to assess candidate-level donor engagement volume.
- Qty Donations CommitteeName Year (QtyDonCommYr): Total number of contributions directed to each committee per election year, capturing organizational-level donor frequency.
- Qty Donations Year (QtyDonYr): Total number of contributions made per election year. This variable supports the analysis of the geographical intensity of donor activity.

Categorical variables, including ContTypeXContrType, ContributorName, CandidateName, and CommitteeName, were encoded using label encoding to ensure compatibility with machine learning algorithms.

The dataset was then partitioned into 80% for training and 20% for testing, in line with standard modeling practices to preserve generalizability. Feature selection was subsequently carried out using a combination of model-based importance scores derived from XGBoost and correlation analysis. This approach allowed for the removal of redundant variables and the

retention of the most informative features, ultimately enhancing the model's predictive performance.

Finally, numerical features were normalized using Standard Scaler to ensure consistent scaling prior to model training. Table 4 is the resulting dataset structure, which reflects a balance between data richness and cleanliness, providing a robust foundation for downstream predictive modeling.

Table 4 - Summary statistics for the target variable and features (N = 6,411)

Feature Name	Mean	St. Dev	Min.	25%	50%	75%	Max.	Skew.	Kurt.	Corr. with Amount
ElectionYear	2,014.94	1.69	2,010	2,015	2,015	2,015	2,024	0.86	7.73	-0.04
Latitude	38.90	0	38.89	38.9	38.90	38.91	38.92	1.11	-0.01	0.16
Longitude	-77.03	0.01	-77.08	-77.03	-77.03	-77.03	-77.02	-2.11	4.54	-0.18
ContTypeXContrType	70.87	25.95	1	53	88	88	88	-1.31	0.57	-0.13
ContributorName	44,822.80	21,197.77	427	25,809	48,336	58,783	83,640	-0.33	-0.81	-0.12
CandidateName	178.01	71.52	4	165	165	165	485	2.12	6.07	0.03
CommitteeName	591.70	191.08	8	551.50	696	696	747	-1.66	1.42	-0.19
DistWh	0.93	0.68	0.10	0.50	0.50	1.30	4.30	1.78	2.8	0.21
QtyDonCandYr	5,190.52	2,636.96	1	3,001	6,835	6,835	6,835	-1.08	-0.7	-0.18
QtyDonCommYr	5,179.44	2,644.51	1	3,001	6,835	6,835	6,835	-1.07	-0.71	-0.18
QtyDonYr	3,805.50	1,477.14	163	2,797	4,698	4,698	4,698	-1.27	-0.03	-0.18
AmtCandYr	213,492.57	346,297.02	10	128,114	128,114	128,114	2,294,601	4.43	20.79	0.1
AmtCommYr	206,967.71	323,808.57	10	128,114	128,114	128,11,4	2,294,601	4.44	21.22	0.1
AmtYr	645,954.16	814,114.84	40,450	227,928	227,928	564,518	2,654,699	1.75	1.44	0.17
AmtContTypeXContrTypeYr	508,638.18	846,122.55	2,365	98,488	98,488	190,774	2,812,111	1.79	1.54	0.16
Qty25\$DonYr	2,740.26	793.33	163	2,872	3,128	3,128	3,128	-1.98	2.49	-0.15
Qty50\$DonYr	1,325.81	912.75	211	898	898	1,103	3,650	1.82	1.83	0.15
Qty100\$DonYr	1,732.70	1,541.23	427	959	959	1,644	5,588	1.76	1.53	0.16
Qty250\$DonYr	757.98	774.44	129	359	359	882	2,704	1.77	1.61	0.17
Vap	1,118.29	603.42	359	791	791	1,127	2,951	1.66	1.43	0.15
MedAge	34.49	3.29	19.70	33.10	35.9	35.90	55.10	-1.16	3.77	-0.11
MedInc	95,887.73	20,990.68	11,375	90,625	90,625	90,645	230,978	1.76	5.01	0.14
UnempRate	3.58	1.17	0	3.50	3.5	3.5	13.50	2.62	14.36	0.04
BachelorHigher	80.33	7.61	37	79.60	79.6	81	96.10	-1.49	6.08	0.05
AvgHhSize	1.52	0.27	1.23	1.38	1.38	1.52	2.58	2.02	2.83	0.2
ForeignBorn	515.21	188.38	44	512	512	512	1,630	1.58	6.3	0.07
MedHomeVal	489,071.75	194,240.24	236,200	386,200	386,200	503,400	1,626,300	2.26	4.54	0.19
PubTransit	664.92	295.57	77	538	538	653	1,634	2.02	3.12	0.11
Amount	17.83	8.05	1	10	21	25	25	-0.50	-1.39	1

3.2 METHODS

This section outlines the analytical methods used to evaluate the dataset. The chosen techniques are designed to enhance both the accuracy and interpretability of the model's outputs. We employ XGBoost as the primary modelling approach due to its robust performance on structured data, scalability, and ability to capture complex nonlinear relationships. Widely adopted in real-world financial machine learning applications, XGBoost has proven effective in delivering accurate and reliable predictions across various domains, including credit scoring, fraud detection, and investment forecasting (Li & Xu, 2023), making it especially well-suited for modeling political contribution patterns. To optimize model performance, we leverage Optuna for hyperparameter tuning, ensuring an adaptive and efficient search for the optimal configuration. This automated optimization process enhances the model's accuracy, stability, and generalizability. To comprehensively assess model effectiveness, we utilize a set of performance metrics, including MAE, MAPE, and RMSE, that provide a multidimensional evaluation of predictive accuracy, consistency, and generalization. Finally, we incorporate SHAP (SHapley Additive Explanations) to enhance the transparency and interpretability of the model's decision-making process. SHAP provides a detailed understanding of how individual features influence predictions, supporting trust, insight, and actionable analysis. This analytical framework underscores the role of Explainable AI (XAI) and open data in addressing urban challenges, while emphasizing the necessity of transparency in AI-driven analytics.

3.2.1 XGBOOST FOR PREDICTING POLITICAL CONTRIBUTIONS

Political contributions in Washington, D.C., are influenced by a complex interplay of socioeconomic, geographic, and donor-specific factors. These patterns are inherently non-linear and shaped by interactions between features such as income, donor identity and type, election cycle, and spatial proximity to political centers. For example, the influence of high-income contributors may vary drastically depending on the election year or the recipient committee. Such complexities are poorly captured by traditional regression models, which assume linear and additive relationships. To address this, we adopt XGBoost (eXtreme Gradient Boosting) as our predictive model to estimate campaign contribution amounts (Amount). XGBoost is well-suited to financial prediction tasks due to its ability to capture non-linear interactions, manage multicollinearity, and maintain high accuracy even in the presence of imbalanced or noisy data (Li & Xu, 2023). It also efficiently handles large tabular datasets like ours, which include rich contextual and quantitative features such as DistWh, MedInc, UnempRate, and QtyDonCommYr. What makes XGBoost especially effective is its regularized learning objective, which balances predictive accuracy with model complexity. The objective function combines a differentiable loss function l with a regularization term Ω , formally defined in Equation (1):

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(\hat{y}_i, y_i) + \sum_{k=1}^K \Omega(f_k) \quad (1)$$

where:

- $l(\hat{y}_i, y_i)$ measures the error between predicted and actual contributions;
- $\Omega(f_k) = \gamma T + \frac{\lambda}{2} ||w||^2$ penalizes model complexity by limiting the number of leaves in each tree and the magnitude of the leaf weights;
- γ, λ : regularization parameters that help reduce overfitting.

To fine-tune performance, we utilize Optuna, a state-of-the-art hyperparameter optimization library, to adjust critical XGBoost parameters, including learning rate, tree depth, number of estimators, and L2 regularization. This ensures efficient convergence and improved generalization. Finally, to interpret the model and ensure transparency, we apply SHAP (SHapley Additive Explanations) values, which quantify the contribution of each feature to the predicted donation amount (Gebreyesus et al., 2024). This allows us to identify the most influential factors driving political giving and supports more informed, data-driven strategies in campaign finance.

3.2.2 CROSS-VALIDATION STRATEGY

To ensure the robustness and generalizability of our predictive model, we employ k-fold cross-validation (CV), a resampling technique widely used in machine learning to assess model performance while mitigating issues related to data variability and overfitting (Meenu Bhagat & Dr. Brijesh Bakariya, 2025). This approach partitions the training dataset into k subsets (or "folds") and iteratively trains the model on k-1 folds while validating on the remaining fold. The process repeats k times, with each fold serving as the validation set once, and the overall performance is averaged across all iterations. Mathematically, the cross-validation error is computed as in Equation (2):

$$CV_{\text{error}} = \frac{1}{k} \sum_{i=1}^k MAE(M_i) \quad (2)$$

where:

- k is the number of folds (in our case, $k = 5$)
- M_i represents the model trained on the i -th fold.
- $MAE(M_i)$ is the Mean Absolute Error (MAE) computed for that fold.
- CV_{error} is the final averaged error across all folds.

For this study, we used 5-fold cross-validation with the KFold method from `sklearn.model_selection`, which ensures:

- Shuffling of data before splitting, reducing bias in fold selection;
- Consistent random seed (42) for reproducibility;
- Balanced validation, where every observation is used in training and validation at least once;

The XGBoost model is trained and evaluated across the five cross-validation folds. For each fold:

- The dataset is split into training and validation sets;
- The model is trained on the training set using predefined hyperparameters;
- Predictions are made on the validation set;
- The performance metric Mean Absolute Error (MAE), is computed and stored.

Once all five iterations are complete, we compute the average MAE across folds, providing a stable assessment of the model's expected error. This strategy ensures that the model is not overly dependent on any particular subset of the data, thereby minimizing the risk of overfitting. By leveraging cross-validation, we ensure that the trained model generalizes well to unseen data, improving reliability when deployed for real-world predictions of political contributions in Washington, D.C.

3.2.3 HYPERPARAMETER TUNING

Fine-tuning hyperparameters is a crucial step in machine learning, as it directly influences a model's accuracy, generalization, and computational efficiency. In this study, we employ Optuna⁷, an advanced and automated hyperparameter optimization framework (Akiba et al., 2019), to enhance the predictive performance of our XGBoost model. Optuna is widely recognized for its efficiency and flexibility, supporting a range of optimization techniques that streamline the tuning process.

Optuna dynamically searches for the optimal set of hyperparameters by leveraging state-of-the-art optimization strategies, including Covariance Matrix Adaptation Evolution Strategy (CMA-ES), Tree-structured Parzen Estimator (TPE), Random Search, and Grid Search. For this study, we utilize TPE, Optuna's default sampler, which is a Bayesian optimization method. Unlike traditional grid or random search techniques, TPE learns from past evaluations, refining the search space by prioritizing promising hyperparameter regions. This adaptive approach allows for faster convergence, making the tuning process more efficient. A key advantage of Optuna is its pruning mechanism, which discards underperforming trials early in the process. This strategy prevents unnecessary resource allocation to unpromising configurations, significantly reducing computational time while improving overall optimization efficiency.

⁷ Optuna (<https://optuna.org/>)

Additionally, Optuna's built-in visualization tools allow us to analyse trial progress, understand hyperparameter relationships, and identify trends that contribute to model performance.

Given the complexity of our dataset and the need for accurate prediction of campaign contributions (Amount), Optuna plays a vital role in fine-tuning key XGBoost hyperparameters. These include:

- `learning_rate`, which controls the step size during model updates;
- `max_depth`, which determines the depth of individual trees and affects model complexity;
- `n_estimators` (number of boosting rounds);
- `reg_lambda` (L2 regularization), which helps prevent overfitting. These parameters interact to balance model bias and variance, ensuring a robust predictive model that generalizes well to unseen data.

These hyperparameters interact to balance bias and variance, ensuring that the resulting XGBoost model generalizes well to unseen data. Moreover, Optuna supports parallel trials, allowing multiple evaluations to run concurrently across computational resources. This parallelization significantly accelerates the hyperparameter tuning process, especially for computationally intensive models like XGBoost, by distributing the search load efficiently.

By leveraging Optuna, our study benefits from an automated and scalable hyperparameter tuning process, ensuring that the XGBoost model achieves optimal performance in predicting political contributions in Washington, D.C. This approach not only enhances predictive accuracy but also provides a robust, adaptable framework for future applications in machine learning-driven political analysis.

3.2.4 EVALUATION METRICS

Assessing model performance is a critical aspect of machine learning, particularly in predictive modeling, where selecting appropriate evaluation metrics ensures a comprehensive understanding of model accuracy, reliability, and generalization capability. Given that our study focuses on predicting political campaign contributions (Amount) using XGBoost, we employ a set of regression-based evaluation metrics that provide both absolute and relative error assessments. The choice of these metrics aligns with the model's optimization strategy, ensuring consistency between the objective function used during training and the criteria for model evaluation.

For this study, we prioritize Mean Absolute Error (MAE) as our primary evaluation metric, as it provides an intuitive and direct measure of average prediction error. Since our XGBoost model was optimized using MAE through Optuna, using the same metric for evaluation ensures coherence between model training and assessment. MAE is particularly valuable in this context because it treats all prediction errors equally without disproportionately

penalizing large deviations, making it more robust for skewed distributions, a common trait in political contribution data where a small number of donors contribute disproportionately large amounts. The Mean Absolute Error is formally defined in Equation (3):

$$MAE(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

where y_i represents actual contribution amounts, $\hat{y}_i$ is the predicted value, and n is the total number of observations.

While MAE provides a straightforward measure of error magnitude, relying exclusively on a single metric may not fully capture all aspects of model performance. Therefore, we complement MAE with three additional evaluation metrics: Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2 Score). Each of these metrics highlights different aspects of model accuracy and error distribution, ensuring a multifaceted evaluation of our predictive model. MAPE is a scale-independent metric that quantifies prediction errors as a percentage of the actual values, making it useful for understanding the relative magnitude of errors. It is particularly effective for comparing errors across different scales of contribution amounts. The formula for Mean Absolute Percentage Error (MAPE) is given by Equation (4):

$$MAPE(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (4)$$

On the other hand, RMSE provides insight into how large errors impact model performance by squaring the residuals before averaging them. This means that larger errors are given more weight, making RMSE particularly sensitive to outliers. This property is beneficial in cases where larger deviations from actual values must be minimized. RMSE is defined by Equation (5):

$$RMSE(y, \hat{y}) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

While error metrics quantify the discrepancy between actual and predicted values, the Coefficient of Determination (R^2 Score) assesses the proportion of variance in the dependent variable that is explained by the independent variables. It indicates how well the model captures variability in contribution amounts. An R^2 score close to 1 suggests that the model explains most of the variance in the data, whereas a value near 0 implies poor predictive power. The formula for R^2 is defined by Equation (6):

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

By employing this diverse set of evaluation metrics, we ensure a comprehensive and well-rounded assessment of our predictive model. The combination of MAE, MAPE, and RMSE provides insight into both absolute and relative error magnitudes, sensitivity to extreme values, and overall model fit. This framework enables us to accurately evaluate the effectiveness of our XGBoost model in forecasting political campaign contributions, ensuring that the predictions are reliable and interpretable for stakeholders analysing campaign finance patterns in Washington, D.C.

3.2.5 EXPLAINABLE ARTIFICIAL INTELLIGENCE

As machine learning systems continue to gain prominence across various domains, their decision-making processes must be transparent and interpretable, particularly in applications where fairness, accountability, and trust are crucial. Predictive models based on gradient boosting algorithms, such as XGBoost, are often considered “black-box” systems, meaning their internal logic and feature contributions are not readily interpretable. This lack of transparency presents challenges, as users, policymakers, and stakeholders require a clear understanding of how these models derive their predictions. The growing adoption of Explainable Artificial Intelligence (XAI) aims to bridge this gap by enhancing the interpretability of AI-driven decision-making without compromising predictive performance.

Historically, AI advancements have focused on maximizing predictive accuracy, often at the expense of interpretability. However, as machine learning applications expand into critical areas such as political finance analysis, economic forecasting, and policymaking, there is a pressing need for models to offer interpretable and justifiable predictions. XAI methods enable users to gain insights into the rationale behind model outputs, thereby fostering trust and transparency. In this study, we integrate SHAP (SHapley Additive Explanations) as an XAI technique to interpret our XGBoost model, which predicts political campaign contributions (Amount) in Washington, D.C.

SHAP is a powerful interpretability framework rooted in cooperative game theory, designed to quantify the contribution of each feature to a model’s predictions. Unlike traditional feature importance scores, SHAP values consider all possible feature combinations to determine how each input variable influences the predicted outcome. By assigning Shapley values to features, SHAP provides a mathematically sound method for distributing prediction impact fairly among all contributing variables. The SHAP value formula is given by Equation (7):

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (7)$$

where $\phi_i(v)$ represents the SHAP value for feature i , N is the set of all features, S is a subset of features excluding i , $v(S)$ is the model prediction based on subset S , $v(S \cup \{i\})$ and is the prediction including feature i . This formula ensures a fair distribution of predictive influence across all features.

In the context of XGBoost, SHAP enables the interpretation of how individual features, such as contributor identity, committee and candidate names, contribution volume and frequency, as well as socioeconomic and demographic indicators, predict donation amounts. Positive SHAP values indicate that a particular feature increases the predicted contribution, whereas negative values suggest a decreasing effect. The magnitude of SHAP values represents the strength of a feature's contribution, allowing for the comparison of the relative importance of different variables.

One of the major advantages of SHAP is its local and global interpretability. At a local level, SHAP values explain individual predictions, revealing why a particular donor's contribution was estimated at a specific amount. At a global level, SHAP provides an aggregated view of feature importance across the entire dataset, highlighting which factors consistently drive political donations. This dual capability is particularly valuable in political campaign analysis, as it enables data-driven decision-making for fundraising strategies while maintaining model transparency.

By leveraging SHAP within XGBoost, we ensure that our predictive model remains not only accurate but also interpretable. This enhances the trustworthiness of AI-driven political finance analysis, allowing policymakers, researchers, and campaign managers to make informed decisions based on transparent and explainable machine learning insights. The integration of Explainable AI in our study highlights the importance of interpretability in complex predictive models, underscoring the need to balance predictive power with accountability and fairness in AI-driven analytics.

4. RESULTS

This chapter presents the results obtained from a comprehensive analysis focused on predicting micro-contributions to political campaigns in Washington, D.C., using XGBoost as the final and sole modeling approach. XGBoost was selected due to its proven effectiveness in handling the specific challenges posed by the dataset, namely, the right-skewed distribution of the target variable (Amount), the presence of complex feature interactions, and the high dimensionality of the input space. The dataset integrates a wide range of feature types, including spatial proximity measures, socioeconomic indicators, and donor-specific attributes. XGBoost's ability to naturally model non-linear relationships and manage heterogeneous data types makes it particularly well-suited to this context. Moreover, its built-in regularization techniques help control overfitting, enhancing the model's generalization performance across varied subsets of the data. This approach enabled a robust and interpretable analysis of model performance, offering valuable insights into the key drivers of political contributions in Washington, D.C..

Following the selection and optimization of the XGBoost model, a unified predictive framework was developed, trained, and evaluated using an enriched set of features derived from multiple open data sources. This approach enabled a detailed examination of model performance, providing a comprehensive understanding of the key factors influencing political contributions. Subsequently, we employ SHAP (SHapley Additive exPlanations) analysis to interpret the predictions made by the unified model, offering deep insights into feature importance, feature interactions, and their respective contributions to the final predictions.

In this section, we present the results obtained using Python (version 3.11) in a Google Colab environment⁸. Data manipulation and analysis were carried out using pandas (version 2.2.2), numpy (version 2.0.2), and scikit-learn (version 1.6.1). Machine learning tasks were implemented using xgboost (version 3.0.0) and tensorflow (version 2.18.0). For hyperparameter optimization, we employed optuna (version 4.3.0) in conjunction with scipy (version 1.14.1). Visualization and plotting were facilitated by matplotlib (version 3.10.0), seaborn (version 0.13.2), and plotly (version 5.24.1). Static image export was supported via kaleido (version 0.2.1). Model explanation and interpretability were enhanced using shap (version 0.47.1). Intermediate storage and object serialization were handled with joblib (version 1.4.2). All workflows were executed within the Colab-hosted virtual machine, utilizing high-RAM runtime and NVIDIA T4 GPU acceleration when required.

⁸ Google Colab (<https://colab.research.google.com/>)

4.1 ANALYSIS OF THE PREDICTIVE MODEL

The XGBoost model was evaluated using a comprehensive suite of regression metrics, including Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and the R^2 coefficient. These metrics were applied across both the training and testing datasets to assess the model's accuracy, consistency, and generalization to unseen data. Notably, the model achieved a Mean Absolute Error of 1.56, corresponding to an average error of approximately 6.2% in predicting contribution amounts. This level of precision is particularly strong given the inherent complexity and variability of political donation behavior, and it demonstrates the effectiveness of XGBoost in modeling such financial outcomes.

4.2 MODEL TRAINING AND OPTIMIZATION

Our analysis employs the XGBoost model, trained on an 80-20 train-test split. The model undergoes independent training on the training dataset, with hyperparameter optimization conducted using Optuna. Within Optuna, the objective function utilizes 5-fold cross-validation to compute the model's Mean Absolute Error (MAE) based on the best hyperparameters found. The data are randomized before fold splitting, ensuring that each fold is a representative mix of the whole dataset, thereby improving the robustness and reliability of the cross-validation process and preventing any order-based biases. A fixed random seed of 42 was set during shuffling to ensure reproducibility.

The model is trained and evaluated during every cross-validation fold, with MAE calculated at each step. Optuna intelligently explores the hyperparameter space, reporting intermediate results and pruning underperforming trials to accelerate the optimization process. Early stopping is also employed to prevent overfitting during training. This cross-validation procedure, dividing the dataset into five subsets (folds), ensures that every data point is used for both training and validation, and the average MAE across folds is used as the optimization criterion.

Optuna's objective is to identify the combination of hyperparameters that minimizes this average MAE. This iterative process spans 100 trials. For each trial, the model is trained with a different set of hyperparameters to determine which yields the best MAE score. Once the best-performing hyperparameter set is identified, the final XGBoost model is retrained on the entire training dataset to solidify its learned patterns.

The final model is then evaluated on the unseen test dataset. Predictions are made, and model performance is assessed using multiple metrics, including MAE, MAPE, RMSE, and R^2 , which are computed on the test set. Residuals are additionally analyzed to diagnose potential prediction issues and validate model assumptions.

This predictive analytics pipeline, which involves model training, hyperparameter tuning, and cross-validation, ensures a rigorous and reproducible modeling process designed to maximize predictive performance. Additionally, a fine-tuning process was conducted with multiple runs to refine the hyperparameter search space iteratively. This involved adjusting search ranges based on previously best-found values and considering the sensitivity of each hyperparameter to model performance. Table 5 presents the hyperparameters used in the XGBoost model, which enabled balancing model complexity with generalization to unseen data, thereby achieving an optimal trade-off between predictive power and overfitting prevention.

Table 5 - Optimal hyperparameters of the XGBoost model (best value and search range)

Hyperparameter	Description	Type	Model
n_estimators	The number of gradient boosted trees. Equivalent to the number of boosting rounds. Too many trees can lead to overfitting by capturing more noise.	Integer	2400 (1500, 2500)
max_depth	Maximum tree depth for base learners. Controls how deep each tree is allowed to grow during any boosting round. Deeper trees can capture more noise	Integer	12 (9, 12)
learning_rate	Step-size shrinkage is used to prevent overfitting. It scales the contribution of each tree by a factor between 0 and 1.	Float	0.0117 (0.005, 0.012)
subsample	The proportion of data samples utilized to train each base learner in the model. Values less than 1 make the algorithm more conservative and prevent overfitting.	Float	0.85 (0.6, 0.85)
colsample_bytree	The fraction of features to be used for each tree. A value of 0.6 means that 60% of features are used to train each tree.	Float	0.6 (0.6, 0.85)
gamma	Required minimal reduction in loss to further split a leaf node in the tree. A larger gamma simplifies the model, preventing overfitting.	Float	3.5 (1.0, 5.0)
reg_lambda	L2 regularization term (Lasso) on weights. It is used to avoid overfitting by penalizing large weights.	Float	6.5 (6.0, 12.0)
reg_alpha	L1 regularization term (Ridge) on weights. It encourages sparsity in the final model representation, allowing some features to be entirely ignored.	Float	5.0 (2.0, 6.0)
min_child_height	Minimum sum of instance weight (hessian) required in a child node. Higher values help prevent overfitting by avoiding learning from overly specific data patterns.	Integer	7 (5, 15)
max_leaves	Maximum number of leaves per tree. Helps control overfitting by restricting tree depth in a more direct and interpretable way than max_depth.	Integer	102 (80, 130)

In general, the Model is engineered for precision and adaptability, although it does not align with the historical contributions to political campaigns in Washington, D.C. Although it combines detailed pattern detection with strategies to ensure reliability and generalizability, the model struggles to predict very high and very low contributions.

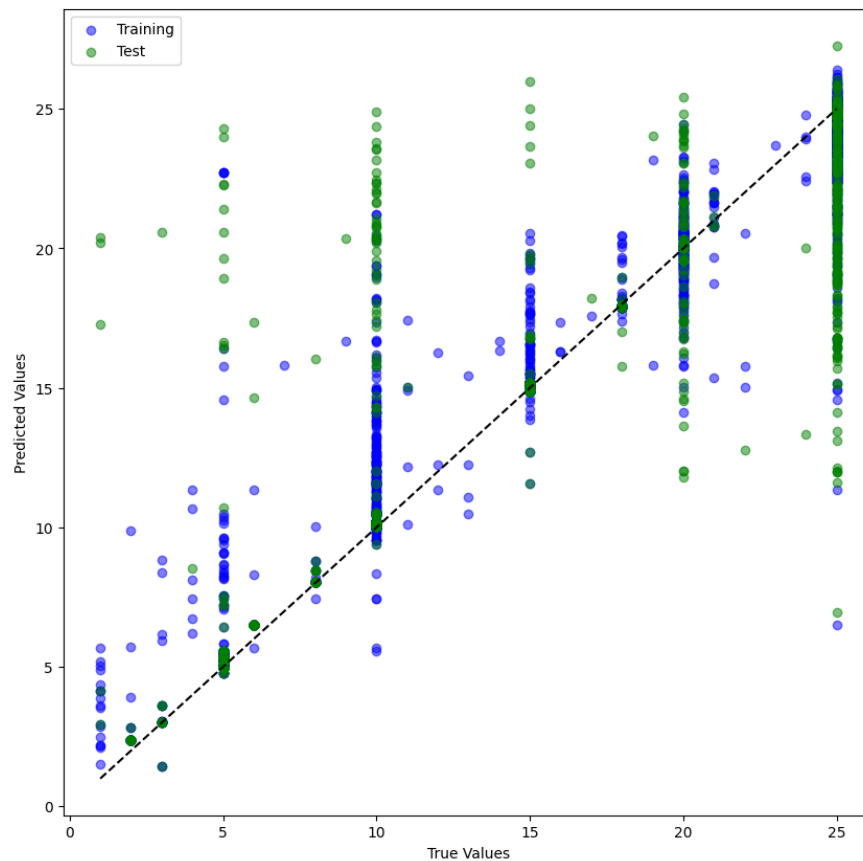


Figure 4 - Prediction error plot for campaign contribution amounts

Figure 4 presents the prediction error plot for the contribution amount (Amount). The model demonstrates relatively strong alignment between predicted and actual values in the mid-range of the target variable, as shown by the dense concentration of points clustered around the identity line. This suggests the model can reliably estimate contributions that fall within the central portion of the distribution. However, the plot also reveals performance limitations at both ends of the value spectrum. For lower contribution amounts, particularly those below \$10, the model frequently overestimates, evidenced by a noticeable vertical dispersion of points above the diagonal in the lower-left quadrant of the graph. Conversely, for higher true values (approaching and exceeding \$20), the predictions increasingly fall below the identity line, indicating a tendency to underpredict large contributions.

Additionally, a visual distinction emerges between training and test predictions. The training set displays a tighter alignment around the identity line, reflecting a closer fit to the data it was trained on. In contrast, test predictions exhibit greater spread, particularly at the

extremes, highlighting the model's challenges in generalizing to unseen data with highly skewed or rare target values. Overall, the figure highlights the model's solid performance in the mid-range of micro-contribution amounts, while also revealing systematic overprediction of small contributions and underprediction of high ones.

Figure 5 - Residual plot for predicted campaign donation amounts

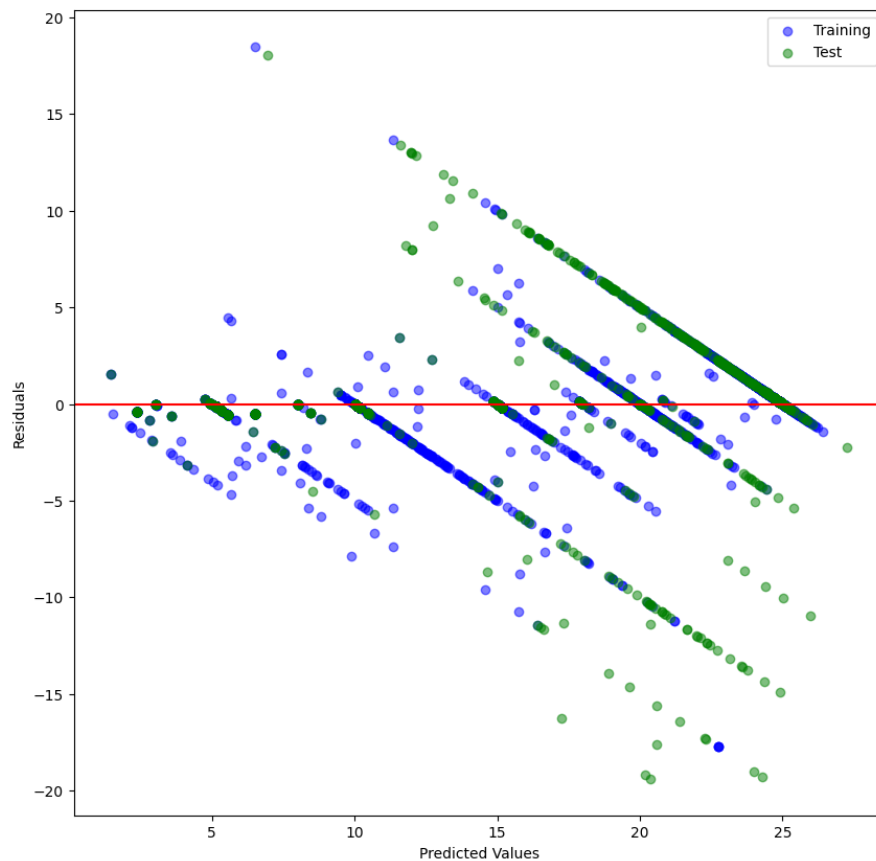


Figure 5 presents the residuals plotted against the predicted values to evaluate the distribution and structure of the model's prediction errors. In an ideal scenario, residuals should appear randomly scattered around the horizontal zero line, indicating that the model has captured the underlying patterns in the data without systematic bias or error trends. However, the plot reveals several structural issues in the residuals that warrant discussion.

A clear increase in error spread emerges: as predicted values increase, residuals tend to spread more widely and skew downward. This indicates that the model increasingly underestimates the true values for larger predicted amounts. In contrast, for smaller predicted amounts, particularly those below \$15, residuals are more symmetrically distributed around zero but still exhibit noticeable variance, indicating challenges in predicting low-value contributions with consistent precision. The residuals also follow a downward sloping band, suggesting that errors are not randomly distributed but rather exhibit a systematic bias, particularly underprediction for high predicted values and overprediction for some mid-range values. This shape reflects difficulty in modeling the extremes of the distribution, which is consistent with the findings in the prediction error plot.

The contrast between training and test points further reinforces this interpretation. Training data residuals are more concentrated, while test residuals are visibly more dispersed, especially at the upper end of the predicted range. This suggests reduced generalization capacity for unseen data, particularly for higher-value donations.

In sum, the residual pattern indicates that although the model captures much of the central structure in the data, it struggles to generalize effectively across the full range of contribution values. These challenges likely stem from the complex, imbalanced, and highly skewed nature of political donation data, which may still be influenced by unobserved variables such as campaign context, donor motivations, or external political events.

4.3 MODEL EVALUATION

After training and hyperparameter optimization, we evaluated the performance of the model, using various metrics to do the assessment: MAE, MAPE, RMSE, and R^2 computed for the test set.

The calculated performance metrics for the model in Table 6 provide quantitative insights into its predictive capabilities for political contributions in Washington, D.C. The performance metrics calculated on the test set estimate how the model will likely perform in real-world scenarios, thereby providing insight into the model’s value.

Table 6 - XGBoost performance

Dataset		Train				Test			
Metrics	MAE	MAPE	RMSE	R^2		MAE	MAPE	RMSE	R^2
Performance	0.58	5.8	1.4	0.96		1.56	17.07	3.56	0.80

After training and hyperparameter optimization, the XGBoost model was evaluated using a comprehensive set of regression metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), calculated for both training and test datasets. These performance indicators, summarized in Table 5, offer a well-rounded assessment of the model’s ability to predict political contributions in Washington, D.C., under realistic conditions. The model demonstrated exceptionally strong performance on the training data, achieving an MAE of 0.58, an RMSE of 1.40, and an R^2 of 0.96, indicating that the model effectively captures complex relationships within the feature space. While this level of training accuracy is notable, some divergence is observed on the test set, a common and expected outcome when modeling real-world behavioral data that is both noisy and imbalanced. On the test set, the model achieved an MAE of 1.56 and an RMSE of 3.56, with an R^2 of 0.80, demonstrating a

solid ability to generalize and retain predictive power even under more volatile and unseen conditions. The MAPE of 17.07%, though higher than in training, reflects the influence of small dollar contributions, where small absolute errors can disproportionately inflate percentage-based metrics. Despite this, the magnitude of error remains within a reasonable range, particularly for applications focused on strategic insights, contribution pattern analysis, and targeted outreach.

Overall, the results suggest that while the model exhibits signs of a tight training fit, it maintains a strong generalization capacity across diverse input conditions. The integration of enriched features, including open data indicators, spatial proximity metrics, and donor-specific characteristics, enabled the model to uncover meaningful patterns and deliver reliable, scalable predictions. These outcomes underscore the practical value of XGBoost in campaign analytics, offering actionable insights for political fundraising strategies and electoral planning.

4.4 MODEL INTERPRETABILITY

To help understand the model, the final phase of this study involves an interpretability analysis using SHAP (SHapley Additive Explanations) to explain the predictions made by the XGBoost model. This approach aims to provide transparency into how the model arrives at its predictions of political contribution amounts, offering insights not only into feature importance but also into the contribution of each variable to individual predictions. To conduct this analysis, we employed SHAP's TreeExplainer, which is specifically optimized for tree-based models such as XGBoost. SHAP values were computed for the test set, allowing us to interpret the model's behavior on unseen data. Each SHAP value quantifies how much a given feature contributes, either positively or negatively, to a particular prediction. This granular interpretability helps reveal which features the model relies on most heavily and how their influence varies across individual cases. The SHAP analysis also enables a better understanding of the directionality and magnitude of each feature's impact. For example, some features consistently increase predicted donation amounts, while others tend to reduce them.

This level of interpretability is especially valuable in politically sensitive applications, where understanding the rationale behind model outputs is essential for accountability and informed decision-making. Several visualizations were generated to support this interpretive phase, including SHAP summary plots and feature impact distributions. These visual tools⁹ help uncover the global importance of features as well as provide localized explanations for specific predictions.

⁹ SHapley Additive exPlanations (<https://shap.readthedocs.io/>)

Figures 6 and 7 both present key visual outputs from the SHAP analysis, which together offer a comprehensive view of how the model internalizes patterns within the political contribution dataset.

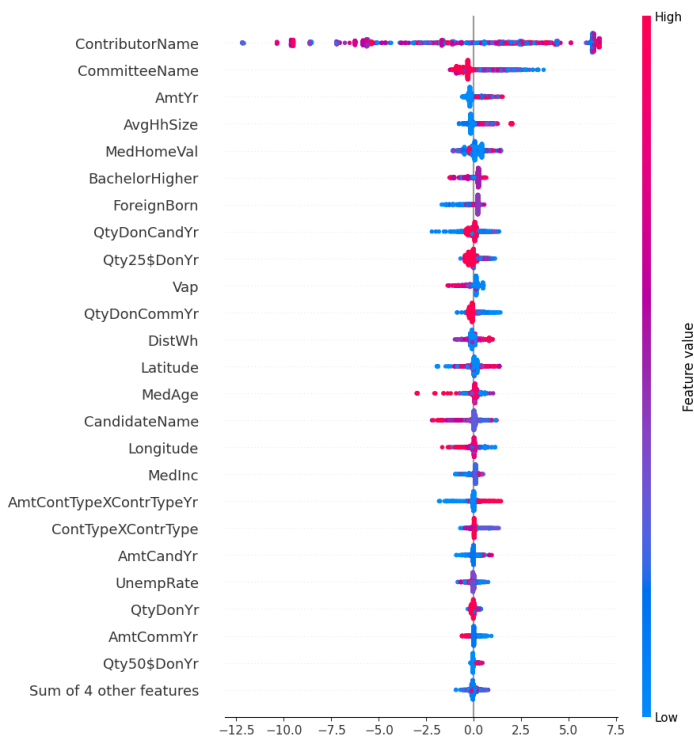


Figure 6 - SHAP Features impact

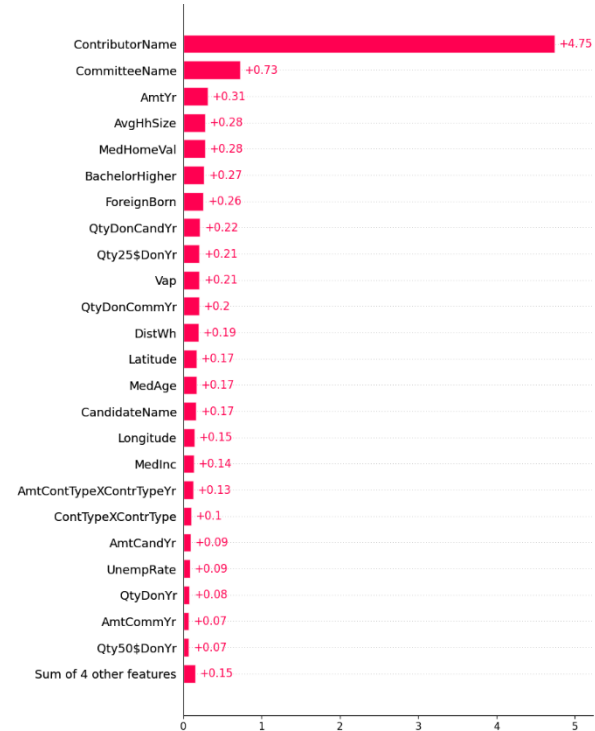


Figure 7 - SHAP Features

To interpret the internal decision-making process of the XGBoost model developed for predicting political contributions in Washington, D.C., this study employs SHAP (SHapley Additive exPlanations). SHAP assigns each feature a contribution value for every prediction, enabling both local (individual-level) and global (dataset-level) interpretability. In political finance modeling, such interpretability is essential not only for identifying key predictive drivers but also for ensuring transparency, fairness, and strategic utility. Figures 6 and 7 present two complementary SHAP visualizations: a summary plot (Figure 6) and a global feature importance bar chart (Figure 7).

The SHAP summary plot (commonly known as a "beeswarm" plot) visualizes the distribution of SHAP values per feature across all instances. Each dot represents one observation's SHAP value for a particular feature, with the x-axis indicating its effect on the predicted contribution amount. Colors reflect feature values (blue = low, pink = high), revealing both the importance of each feature and the heterogeneity of its impact on the target variable.

By contrast, the global SHAP importance plot (Figure 7) displays the average magnitude of SHAP values per feature, thereby ranking them according to their overall contribution to

model predictions. Each bar's length reflects the mean absolute SHAP value, offering a clear summary of which features the model relies on most.

The SHAP analysis reveals a complex ecosystem in which political contributions are shaped by a combination of behavioral, organizational, geographic, and demographic factors. The model underscores the dominance of campaign and contributor-level signals, followed by contextual socioeconomic patterns. Several clusters emerge:

1. **Contributor Identity:** ContributorName stands out as the most impactful feature in the model, with an average SHAP value of +4.75. This variable likely captures strong individual patterns associated with frequent small-dollar donations from highly engaged contributors. Its dominant presence suggests that donor identity is a critical driver of micro-contribution behavior.
2. **Committee and Candidate Identity:** CommitteeName and CandidateName are moderately impactful features in the model, with average SHAP values of +0.73 and +0.17, respectively. These variables likely capture recurring patterns tied to well-known committees and recognizable candidates. Their presence suggests that the identity of organizational and political figures influences micro-donation behavior, especially when certain names consistently attract engagement from small-dollar contributors.
3. **Contribution Volume and Frequency:** AmtYr, QtyDonCandYr, Qty25\$DonYr, QtyDonCommYr, AmtContTypeXContrTypeYr, AmtCandYr, QtyDonYr, AmtCommYr, and Qty50\$DonYr contribute modestly to the model, with SHAP values ranging from +0.31 to +0.07. These features reflect the timing, amount, and type of contributions made across the year, capturing behavioral traits such as donor consistency, campaign targeting, and micro-giving patterns. Their combined presence suggests that, while no single volume or frequency feature dominates, together they help the model contextualize donation behavior, particularly in identifying trends of active participation within the micro-donation space.
4. **Socioeconomic and Demographic Context:** AvgHhSize, MedHomeVal, BachelorHigher, ForeignBorn, Vap, MedAge, and MedInc contribute moderate yet meaningful SHAP values, ranging from +0.28 to +0.14. These features enrich the model by embedding local socioeconomic conditions, with the most impactful being average household size and home value. Characteristics such as educational attainment, immigrant presence, and age composition further shape the micro-donation landscape. While their influence is weaker than individual-level traits, they provide important context for understanding how neighborhood structure affects donor behavior.
5. **Geographic Information:** DistWh, Latitude, and Longitude reflect spatial dimensions of contribution patterns, with SHAP values ranging from +0.15 to +0.19. Although not

primary drivers in the model, these features capture localized variation in donor behavior, potentially acting as proxies for political proximity, neighborhood context, or civic accessibility. Their influence is limited, likely due to the small geographic scope of the analysis (Ward 2, 15.3 km²), but they still provide useful spatial structure to the model.

6. **Contribution Type by Contributor Type:** ContTypeXContrType shows minimal impact in the model, with an average SHAP value of +0.10. This feature, which captures the interaction between donation method and donor type, adds limited predictive value when modeling micro-donations, where variation in contribution channels and donor categories tends to be low.

The SHAP analysis reveals that political contributions in Washington, D.C., are primarily influenced by donor identity, campaign engagement, and historical donation behavior. High-frequency and high-value contributors, along with prominent committees, exert the greatest impact on predicted contribution amounts. In contrast, demographic and geographic factors play a secondary role. These insights reinforce concerns about the outsized influence of a small, active donor class and highlight the need for transparency and reform in political fundraising. SHAP offers a valuable tool for uncovering these dynamics and informing equitable, data-driven policy decisions.

5. DISCUSSION

This chapter comprehensively analyzes and discusses the results obtained, elucidating their significance and implications in the broader real environment of fundraising political campaigns through micro-donations.

5.1 IDENTITY OF CONTRIBUTOR

According to the SHAP analysis, the feature ContributorName exhibits the highest mean absolute SHAP value (4.64), indicating that it is the most influential predictor in the model. This strong average impact highlights the central role played by individual contributor identity in shaping financial contribution behavior, surpassing all other variables, including socioeconomic and geographic factors. The prominence of this feature suggests that the model relies heavily on patterns associated with specific individuals, emphasizing the predictive power encoded in prior donation history, individual-level behavior, and potential relational dynamics. This finding aligns with the conclusions drawn by Dong et al. (2022), who demonstrate that repeated behavior from individuals provides valuable historical data that significantly enhances the accuracy of machine learning models. When a contributor donates multiple times, the model gains access to a richer behavioral profile, enabling it to learn specific trends, preferences, and frequencies in user actions. Such accumulated individual data is particularly effective in enhancing the predictive performance of models that deal with personalized patterns, such as those found in repeat purchase prediction or user-level recommendation systems. In this context, ContributorName serves as a proxy for behavioral continuity, allowing the model to fine-tune predictions based on previously observed actions and enhancing its ability to anticipate future contributions with greater precision.

The SHAP beeswarm and feature importance plots undoubtedly underscore the dominant role of ContributorName in the model's decision-making process. Not only does it exhibit the highest mean absolute SHAP value (+4.64), but it also displays the widest horizontal spread in the beeswarm distribution, revealing considerable variation in its influence on the model's output. This wide dispersion of SHAP values signals that the feature's effect is highly context-dependent and non-linear. In some cases, a contributor's name contributes significantly and positively to the predicted donation amount, while in others, its impact may be neutral or even negative. Such heterogeneity suggests that the model has learned nuanced behavioral patterns specific to individual donors, for instance, recognizing that some contributors consistently donate at the \$25 threshold, while others display erratic or one-time behavior.

5.2 COMMITTEE AND CANDIDATE INFLUENCE ON DONORS' BEHAVIOR

From the results of the SHAP analysis, the features CommitteName and CandidateName emerge as meaningful predictors in the model, yet their relative contributions differ notably.

With a mean absolute SHAP value of +0.73 for CommitteeName compared to +0.17 for CandidateName, it is evident that the identity of the campaign committee carries substantially more weight in shaping the model's output. This disparity highlights the central role that institutional affiliations, more so than individual politicians, play in the structuring of political contributions.

The predictive strength of CommitteeName suggests that donors often direct their contributions based not solely on ideological affinity, but rather on strategic considerations linked to the perceived influence and political leverage of specific committees. This aligns with the findings of Teso (2025), who demonstrated that corporate and interest group donors tend to funnel resources toward committees chaired by or connected to politicians overseeing relevant policy areas. These committees are often seen as gateways to regulatory influence, thus attracting higher and more targeted donations. In this context, CommitteeName functions as a robust proxy for policy impact potential, explaining its dominant role in the model's predictions. While CandidateName does influence the model, its impact is significantly more muted. With a mean absolute SHAP value of only +0.17, the feature contributes meaningfully, but less consistently. Nevertheless, its presence in the top tier of important features confirms earlier findings by Barber et al. (2019), who argued that donors often seek ideological congruence, contributing to candidates whose policy positions closely mirror their own. The inclusion of CandidateName in the model thus captures donor behavior that is personalized and belief-driven, albeit less potent than committee-based strategic giving.

The SHAP beeswarm analysis reinforces this conclusion that CommitteeName exhibits a much broader horizontal dispersion of SHAP values, stretching from strongly negative to prominently positive, indicating substantial heterogeneity in how different committees affect the predicted donation amounts. High SHAP values (in pink) are concentrated strongly around certain committee identities, reinforcing the idea that donors associate some committees with higher strategic returns or greater visibility. Conversely, CandidateName, while also showing a non-trivial SHAP spread, displays a narrower range of influence. Its SHAP values cluster more tightly around zero, with fewer extreme values on either side. This pattern suggests that while candidate identity does affect donation behavior, likely through mechanisms like public recognition, ideological alignment, or electoral competitiveness, these effects are less systematically influential than those tied to committee affiliations.

5.3 AMOUNT DONATED AND CONTRIBUTIONS MADE PER YEAR

The SHAP analysis indicates that the feature AmtYr (total amount contributed per year) emerges as one of the top predictors in the model, with a mean absolute SHAP value of +0.31. The centrality of AmtYr underscores the importance of aggregate yearly donation volumes in determining the model's ability to predict political contribution behavior accurately.

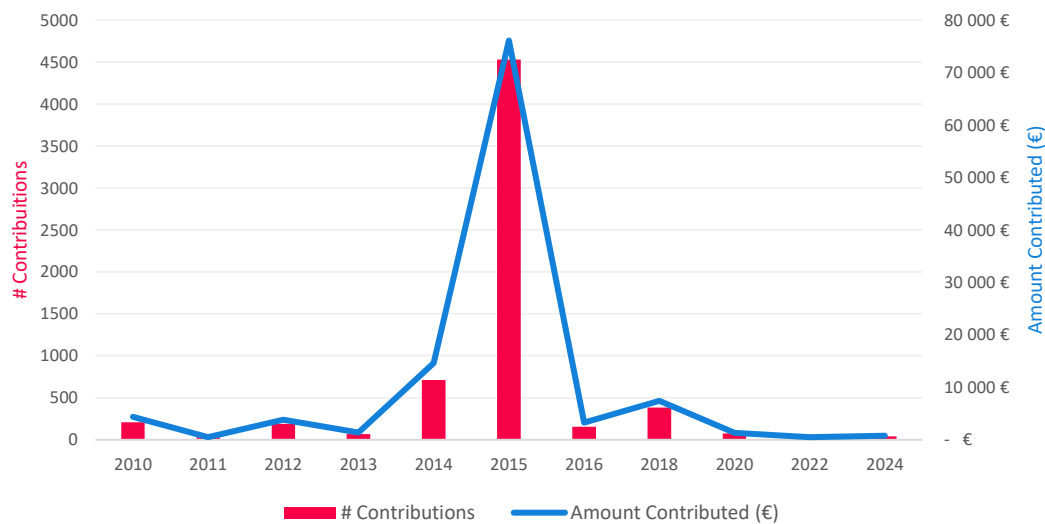


Figure 8 - Trends in Political Contribution Volume and Value in Ward 2 (2010 - 2024)

The predictive strength of AmtYr is best understood through two complementary mechanisms: first, years with higher total donation amounts often reflect more intense political activity and larger contributions. Second, and perhaps more crucially, such years provide the model with a richer dataset to learn from, increasing statistical power and improving the model's ability to generalize. For example, as seen in Figure 8, the year 2015 stands out with over 4,500 contributions totaling €76,100, while a year like 2011 saw only 28 contributions totaling just €500. This stark contrast illustrates why AmtYr carries such predictive weight: high-contribution years, such as 2015, not only reflect elevated donor engagement but also provide the model with significantly more examples to infer meaningful patterns. This interpretation aligns with the findings of Hu (2023), who demonstrated that total value magnitude, such as cumulative sales or transactions in a given period, can substantially improve model accuracy. Larger datasets contribute to better generalization, reduce overfitting, and enhance the model's ability to capture complex relationships, especially when using ensemble methods such as XGBoost, Random Forests, and GBDT. In this context, AmtYr acts as an informational amplifier, enabling the model to refine its predictions by learning from data-rich temporal segments.

The importance of AmtYr is complemented by several related features that reflect contribution behavior and structure. These include:

- QtyDonCandYr (+0.22), indicating the number of contributions made to candidates per year,
- Qty25\$DonYr (+0.21), representing the frequency of 25\$ donations,
- QtyDonCommYr (+0.20), measuring yearly contributions to political committees, and
- AmtContTypeXContrTypeYr (+0.13), capturing the interaction between contribution type and donor type at a yearly level.

Together, these features suggest that not only the total money contributed (AmtYr), but also the frequency and form of giving across targets (candidates, committees, types) play a secondary yet meaningful role in predicting political donation behavior. Other features, although showing less or minimal impact individually, still contribute to subtle patterns in the model. These include:

- AmtCandYr (+0.09), indicating the total amount donated to candidates over a year,
- QtyDonYr (+0.08), representing the overall number of political contributions made annually by each donor,
- AmtCommYr (+0.07), measuring the yearly total of donations directed to political committees
- Qty50\$DonYr (+0.07), capturing the frequency of \$50-level contributions per year.

The SHAP beeswarm plot further substantiates this analysis. AmtYr displays a broad horizontal dispersion, with SHAP values ranging from strongly negative to highly positive, depending on the election year. Higher SHAP values cluster around years like 2014 and 2015, which are associated with greater contribution volumes and higher data density, while lower values are aligned with data-poor years, such as 2011, 2022, or 2024. Donation frequency variables (QtyDon* features) show more constrained horizontal spreads, indicating that their effects are more localized and consistent across observations. Together, these variables suggest that both the total value and behavioral distribution of contributions are key to accurately modeling political giving.

5.4 SOCIOECONOMIC AND DEMOGRAPHIC FACTORS INFLUENCING CONTRIBUTION BEHAVIOR

Another important dimension for predicting political contributions is the set of socioeconomic and demographic variables, which, while exhibiting lower mean SHAP values compared to geolocation and identity-based features, still provide meaningful insight into donor behavior. In this case, features such as Average Household Size (+0.28), Median Home Value (+0.28), Bachelor's degree or Higher (+0.27), Foreign-Born Population (+0.26), and Voting Age Population (VAP) (+0.21) stand out as the most influential within this group.

Meanwhile, features such as Median Age (+0.17), Median Income (+0.14), and Unemployment Rate (+0.09) register a smaller but consistent impact. These variables capture the broader

structural and social context in which donors reside, providing background conditions that shape but do not deterministically drive donation behavior. The SHAP beeswarm plots confirm that these features produce narrower spreads compared to identity-based attributes such as ContributorName or CommitteeName, indicating that their influence is more uniform across the donor population.

For instance, higher average household sizes are modestly associated with increased donation predictions, which may reflect denser family networks or multi-generational household structures that are more common in urban settings, where political discourse and civic engagement are more active.

Median Home Value also exhibits a steady positive contribution to the model, a finding that resonates with the research of Lessem et al. (2023). Their paper demonstrated a causal relationship between rising housing prices and increased political donations, particularly in areas with a Democratic lean. The authors found that ZIP codes with higher home appreciation levels exhibited significantly more donations to Democratic candidates, while Republican candidates saw little to no effect. This dynamic is especially relevant in the context of Washington, D.C., which forms part of the presidential “blue wall” having consistently voted Democratic in every election since 1988. In such environments, rising property values may correlate with both increased wealth and liberal political alignment, reinforcing the predictive value of this feature.

Educational attainment, represented by BachelorHigher, similarly plays a positive role, supporting longstanding theories that higher education correlates with greater political participation. This aligns with findings from Pratiwi et al. (2023), who identified education, income, and age as significant predictors of political engagement across various contexts.

The feature ForeignBorn also emerges as moderately important, though its interpretation is inherently complex and context-dependent. In Washington, D.C., approximately 14% of the population is foreign-born, with nearly 60% of these residents holding a college degree or higher, and over 80% reporting that they speak English well or very well¹⁰. This immigrant profile is notably more educated and linguistically integrated than the national average, where only ~27% of immigrants hold a college degree and ~74% report high English proficiency, according to the U.S. Census Bureau's 2021 American Community Survey. Comparatively, cities like New York City and Los Angeles report that approximately 36% of their populations are foreign-born, while Chicago stands at around 20% and Boston at 25%¹¹. These statistics suggest that D.C.'s immigrant community is both demographically distinct and well-positioned

¹⁰ American Immigration Council (<https://www.americanimmigrationcouncil.org>)

¹¹ U.S. Census Bureau (<https://www.census.gov/>)

for civic participation. The city's proactive stance, encompassing sanctuary city protections, robust legal aid programs, and integration initiatives like the Immigrant Justice Legal Services¹², provides structural support that may enhance immigrant engagement, including political contributions. However, while these factors support the idea that the ForeignBorn variable captures increased immigrant involvement, the interpretation remains multi-faceted. It may also reflect a reactive mobilization among native-born residents, particularly in response to shifts in demographic patterns or increased immigration salience. In this light, the feature could simultaneously signal inclusive political participation and identity-based political reactivity. Thus, ForeignBorn likely acts as a proxy for deeper sociopolitical dynamics, shaped by both welcoming institutional environments and evolving group-based perceptions. While the data supports immigrant-driven engagement in D.C., a dedicated follow-up study would be necessary to disentangle the competing causal mechanisms underlying this SHAP influence. Regardless, its consistent presence in the model affirms the enduring relevance of immigration-related variables in understanding the political economy of campaign finance.

The VAP (Voting Age Population) variable, though less prominent than the features above, still contributes a meaningful signal. A higher VAP may indirectly proxy for denser and more electorally active districts, where political campaigns tend to concentrate fundraising efforts. In contrast, features such as Median Age, Median Income, and Unemployment Rate exhibit lower SHAP impacts, but their presence still contributes to the model's stability. These indicators are known to influence political attitudes and engagement potential, even if their predictive role is more diffuse. As highlighted by Brown-Iannuzzi et al. (2017), socioeconomic status does not straightforwardly translate into political behavior, while it may inform political interest or ideology, these attitudes do not always culminate in actions such as campaign contributions. Even so, although they do not contribute much to the model individually, Hu (2023) noted that predictive performance improves with data richness and feature informativeness, which helps explain why even lower-SHAP socioeconomic features can improve the model's generalizability.

In summary, the socioeconomic and demographic indicators in the model offer a contextual foundation for political behavior, complementing the more dominant identity-based features. Their consistent impact suggests that while donor identity and organizational targeting drive the most variance, structural factors like income, education, housing, and age shape the underlying landscape of political engagement. These variables enhance the model's ability to capture real-world conditions that inform the likelihood and magnitude of political giving, making them essential, if secondary, components of the predictive architecture.

¹² 2025 Immigrant Justice Legal Services [PDF](https://communityaffairs.dc.gov/) on (<https://communityaffairs.dc.gov/>)

5.5 GEOLOCATION

Another important dimension for predicting political contributions is geolocation, as captured by the spatial features DistWh (distance to the White House, in kilometers), Latitude, and Longitude. Although these features do not rank among the top predictors in the current model, with mean SHAP values of +0.19, +0.17, and +0.15, respectively, they still provide meaningful contextual insight into the spatial dynamics of donor behavior. The relatively modest impact of these variables can likely be attributed to the narrow geographic scope of the dataset, which is limited to Ward 2 in Washington, D.C., an area of just 5.9 square miles (approximately 15.3 km²). In such a compact urban zone, geographic variation is naturally constrained, potentially diluting the influence of spatial differentiation on contribution patterns. Nonetheless, the SHAP beeswarm plots reveal that these geolocation features do exhibit heterogeneous effects across observations, suggesting that even within a small area, micro-geographic factors may still exert some influence. Specifically, Latitude shows mild positive SHAP contributions at higher values (in pink), while Longitude tends to exert more influence at lower values, suggesting small but detectable location-based patterns in donor behavior. The feature DistWh, despite its constrained range, continues to show a weak but consistent trend where proximity to the White House is slightly associated with increased predicted contributions, possibly reflecting symbolic or strategic proximity to the federal seat of power. These empirical findings, although limited in scale, are consistent with broader theoretical and empirical research. For instance, Sebold and Mitchell (2014) demonstrated that geographic proximity to political institutions significantly shapes donor behavior. Their study, based on county-level data from the 2004 to 2012 U.S. presidential pre-primary campaigns, identified spatial clustering in campaign financing and argued for the methodological necessity of incorporating geographic data into models of political giving. While the constrained geography of Ward 2 may limit the visibility of such effects in this specific model, the literature suggests that spatial context remains a critical factor when the geographic canvas is broader or more variable.

5.6 TYPE OF CONTRIBUTOR AND METHOD OF CONTRIBUTION

From the results of the SHAP analysis, the interaction feature ContTypeXContrType, which combines the method of contribution (e.g., cash, check, in-kind, credit card) with the type of contributor (e.g., individual, corporation, PAC), displays a modest impact on the model's predictions, with a mean absolute SHAP value of +0.10. This suggests that, within the context of the current study, which focuses primarily on micro and small donations, this feature contributes relatively little to determining the size or likelihood of contributions. The limited influence of ContTypeXContrType may stem from the fact that micro-contributions tend to be more homogeneous in structure, often concentrated among individuals who use standardized, low-variance channels, such as credit cards or small cash/check donations. As a result, the nuanced distinctions between different donor-method pairings exert less variation

on the output, and the model naturally prioritizes features with greater variance or contextual differentiation, such as donor identity or contribution timing.

Nevertheless, the theoretical and empirical literature suggests that this feature may hold greater relevance in models focused on larger-scale political donations. For example, Gelman and Roussanov (2024) and Banker et al. (2021) observed that the contribution method affects donor behavior, noting that cashless mechanisms such as credit cards can encourage overspending or psychologically inflated giving. In parallel, Hessami and Uebelmesser (2016) emphasized that in-kind donations are often perceived as more transparent and less vulnerable to corruption, which can increase donor willingness to contribute higher amounts. In such contexts, combinations like “Check × Corporation” or “Credit × Individual” may exhibit strategic intent, signaling different levels of commitment, regulatory exposure, or campaign targeting. While this level of differentiation may not be as impactful in predicting low-value contributions, it could become a critical feature in upper-tier donation models, where strategic financial behavior, legal structuring, and donor–campaign coordination play larger roles. Its low SHAP value here should not dismiss its theoretical importance but rather contextualize its limited impact within the micro-donation focused scope of this study.

6. CONCLUSIONS

This study effectively demonstrated how explainable machine learning models can be applied to accurately predict political micro-donations in Washington, D.C., while also uncovering the most influential factors driving these contributions, thereby addressing the research question. By fully achieving its research objectives. Through the development and validation of a robust predictive model using XGBoost, optimized with Optuna, the study was able to forecast both the occurrence and amount of micro-donations with high accuracy (Mean Absolute Error of approximately \$1.56 and R^2 of 0.8). The research also identified and analyzed the most influential donor characteristics and donation attributes, and applied SHAP (SHapley Additive exPlanations) to provide a transparent interpretation of the model's predictions. Furthermore, by integrating demographic and geographic open data with campaign finance records, the study generated actionable insights into the contextual factors shaping donation behavior. Overall, the findings demonstrate that explainable machine learning can provide both predictive accuracy and interpretability, offering valuable evidence to inform political strategy, policy reform, and future research in campaign finance.

In terms of the most influential features, the analysis revealed that historical donor behavior, the identities of candidates and committees, and the socioeconomic and demographic context of the donor (sourced from open data) were the most significant predictors. These factors contributed far more to the model's performance than geographic features, which had a limited impact, likely due to the compact and relatively homogeneous spatial characteristics of Ward 2.

The findings of this research carry practical implications. Political organizations can leverage these insights to refine campaign strategies through data-driven approaches, including improved geographic targeting, more tailored messaging, and optimized donor outreach. Moreover, this work promotes transparency and strengthens the fight against corruption in political campaign financing by leveraging explainable machine learning to analyze and predict micro-donation behavior. The use of explainable AI techniques such as SHAP enhances interpretability, transparency, and accountability in political data modeling. By making complex model outputs more interpretable for non-technical stakeholders, these tools strengthen trust and promote more ethical, informed use of machine learning in political decision-making.

By advancing the field of explainable artificial intelligence (XAI) and demonstrating the value of open data, this study contributes to the responsible, informed, and innovative application of machine learning in the political domain. It underscores the importance of collaboration among data scientists, campaign managers, policymakers, and institutions in building intelligent, efficient, and transparent political systems. This research extends the theoretical potential of machine learning in politics by introducing innovative methodologies and

reinforcing the idea that insights derived from machine learning and artificial intelligence are becoming essential for achieving electoral success.

Nonetheless, the study has several limitations that should be acknowledged. First, the model's performance is inherently dependent on the quality and availability of open data, which can vary significantly across different sources and contexts. Open datasets are often subject to missing, incomplete, or biased information, all of which can influence the robustness and generalizability of the model. Second, although the study undertook an exhaustive exploration of relevant predictors through careful feature engineering, there remains room for improvement. Alternative feature construction strategies could potentially yield different or improved model performances, especially with access to more granular or diverse data. Third, the geographic scope of the analysis is limited to Washington D.C.'s Ward 2, which, while politically and economically significant, may not accurately reflect the patterns observed in other wards or jurisdictions. As such, the findings cannot be readily generalized beyond this localized context. Additionally, the study focuses exclusively on micro-donations, thereby excluding larger financial contributions. However, micro-donations are often viewed as proxies for grassroots engagement and may reveal dynamics that also inform broader donation behaviors. Finally, limitations such as the absence of actual receipt dates and the lack of individual-level donor attributes (e.g., age, gender, religious affiliation, and birthplace) may have constrained the model's ability to capture temporal and socio-demographic nuances, which could have further enhanced its predictive accuracy.

To extend the findings, future research should prioritize enhancing the generalizability of the model by testing it across different cities, states, and political environments beyond Washington, D.C., where distinct characteristics, varying regulatory frameworks, and diverse candidate profiles must be considered to ensure broader applicability. In addition, upcoming studies should explore the attributes of candidates and committees, such as ideological affiliation, value propositions, and historical influence, as these factors are likely to significantly shape donor behavior. A particular emphasis should also be placed on datasets with high-quality temporal information, especially the inclusion of receipt dates for contributions, as time-based analysis may uncover cyclical patterns, peaks aligned with key events, or seasonal variations currently hidden from view. Identifying such temporal dynamics could significantly improve the model's predictive accuracy. Further research should also investigate the integration of probabilistic AI models, which have shown promising results in other domains and could complement the existing framework by offering new insights and enhancing predictive capabilities within the political context. Finally, future work should continue to address the ethical dimensions of applying AI in political settings, particularly concerning data privacy, fairness, and equitable access to the benefits of algorithmic systems. As AI becomes increasingly integrated into democratic processes, ensuring its responsible and transparent use will be essential.

Politics permeates every aspect of our daily lives, whether we acknowledge it or not. From the institutions we interact with to the public spaces we occupy, political decisions shape our lives and collective experience. This study represents a step toward demystifying political contributions and making political data science more transparent, impactful, and accessible to everyone.

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